



CTOS Enquiry via XML (ENQWS)

CTOS Credit Report

Technical Specification

Prepared by :

Information Technology (IT) Department



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Table of Contents

1	Purpose	8
2	Introduction	8
3	Overview	9
3.1	Overview of method 'request'	10
3.2	Overview of method 'requestConfirm'	10
3.3	Overview of method 'requestLite'	10
3.4	Overview of method 'requestParty'	11
3.5	Overview of method 'requestPartyConfirm'	11
4	Process Flow	12
4.1	CTOS Credit Report - Individual	12
4.2	CTOS Credit Report - Company/Business/LLP	12
4.3	CTOS Lite Report	14
4.4	CTOS Score Report	14
4.5	CTOS SME Score Report for Company	14
4.6	CTOS SME Score Report for Business	14
5	Authentication	15
5.1	Private Key Generation	16
5.2	Authentication to the API	16
5.2.1	Signed JWT Token Generation	16
5.2.2	Login API	17
5.2.3	Refresh Token API	18
5.2.4	Access token usage	19
6	Method 'requestParty' and 'requestPartyConfirm'	19
6.1	Request XML Format	19
6.2	Response XML Format – Parties	21
6.3	Response XML Format - Entities – instead of Parties	22
7	Method 'request', 'requestConfirm' and 'requestLite'	24
7.1	Request XML Format	24
7.1.1	BATCH Section	25
7.1.2	RECORDS section	26
7.2	Response XML Format	30
7.2.1	ENQUIRY Report	33
7.2.2	HEADER Section	33
7.2.3	SUMMARY Section	34
7.2.4	ENQUIRY Section	40
7.2.5	ENTITIES Section	40
8	Definition of Sections	41
8.1	Section Summary	41
8.2	Section A	43
8.2.1	Individual	43
8.2.2	Business/LLP	45
8.2.3	Company	47
8.3	Section B	55
8.3.1	Enquiry History	55
8.3.2	Individual – Monitoring and Internal List	56
8.3.3	Company – Monitoring and Internal List	56
8.3.4	US Treasury and UN Security Council	57
8.4	Section C	58
8.5	Section D	60
8.6	Section D2	67
8.7	Section D4	67
8.8	Section CCRIS	67

8.9	Section DCHEQS	75
8.10	Section CCRIS SUPPLEMENTARY	75
8.11	Section ANGKASA.....	77
8.12	Section ETR PLUS.....	79
8.13	Section E.....	84
9	XML to HTML Formatting Using Stylesheet	90
10	Integration Connectivity	90
10.1	UAT Environment.....	90
10.2	Production Environment.....	90
11	Summary Section Status Codes	91
11.1	Successful CTOS Report	91
12	List of purposes.....	93
13	Due Diligent Index.....	93
13.1	Due Diligent Index Table	94
13.1.1	Individual	94
13.1.2	Company	95
14	Internal standardization formats	96
14.1	Standardizations of Subject Name.....	96
14.1.1	Company name transformation.....	96
14.1.2	Individual name transformation	96
15	Record Type for Section D	96
16	State Code.....	97
17	Country Code	97
18	Error Message	103

1 Purpose

The purpose of this document is to describe the interface and protocol used between System Integrator (SI) and CTOS ENQWS platform. This document describes the following:

- SI connection to ENQWS
- SI requesting authentication information from ENQWS
- SI requesting information from ENQWS

2 Introduction

This document describes in detail the input and output (request and response) requirements and formats for CTOS Enquiry using XML (Extensible Markup Language) format.

Enquiry using XML format allows better integration between the CTOS ENQWS platform and SI internal systems for instant processing of data.

The XML schema encourages data extraction in the event data extraction is required for storage, processing and backend processing.

In line with standard XML practices, special characters in XML responses are encoded for proper parsing. For instance, the & character is consistently encoded as & throughout all XML responses.

The following conventions are used in specifying the format of the input and output XML format:

- Words and special characters in bold must be supplied as is.
- Characters enclosed in square bracket ([]) denotes variables supplied by user (for input) or generated by the system (for output).

Together with this documentation in the package, included are the following folders:

1.	Appendix	Contains reference documents
2.	Schema	Contains XML schema for both Request and Response (XSD file)
3.	Stylesheet	Contains XSTL stylesheet to transform XML to HTML Report
4.	Samples	Contains sample request XML and response XML
5.	Certificate	Contains CTOS certificates

3 Overview



All APIs provided are web services over SOAP/HTTP(s).

The Web Service Definition Language (WSDL) file can be obtained from

UAT Environment : https://uat-integration.ctos.com.my/ctos_secure/Proxy?wsdl

Production Environment : https://integration.ctos.com.my/ctos_secure/Proxy?wsdl

Data transfer to and from CTOS ENQWS platform will be using Extensible Markup Language (XML).

The web service exposes the following method:

Method Name	Method Description	Applicable Source
request	Webservice method name to retrieve a CTOS Credit Report.	All sources
requestConfirm	Webservice method name to confirm final submission in the event multiple entities are present. This is a conditional call depending on response from "request".	All sources
requestLite	Webservice method name to retrieve CTOS Lite Report.	Only CTOS
requestParty	Webservice method name to retrieve list of Business Interest (Directors, Shareholders, Sole Proprietor or Partners).	Only CCRIS
requestPartyConfirm	Webservice method name to confirm final submission in the event multiple entities are present. This is a conditional call depending on response from "requestParty".	Only CCRIS

The methods above are provided to the SI in order to build a request before submission to CTOS ENQWS platform. The request is based on a straight-through mechanism unless a condition is met where multiple entities is found which then the SI will require confirmation.

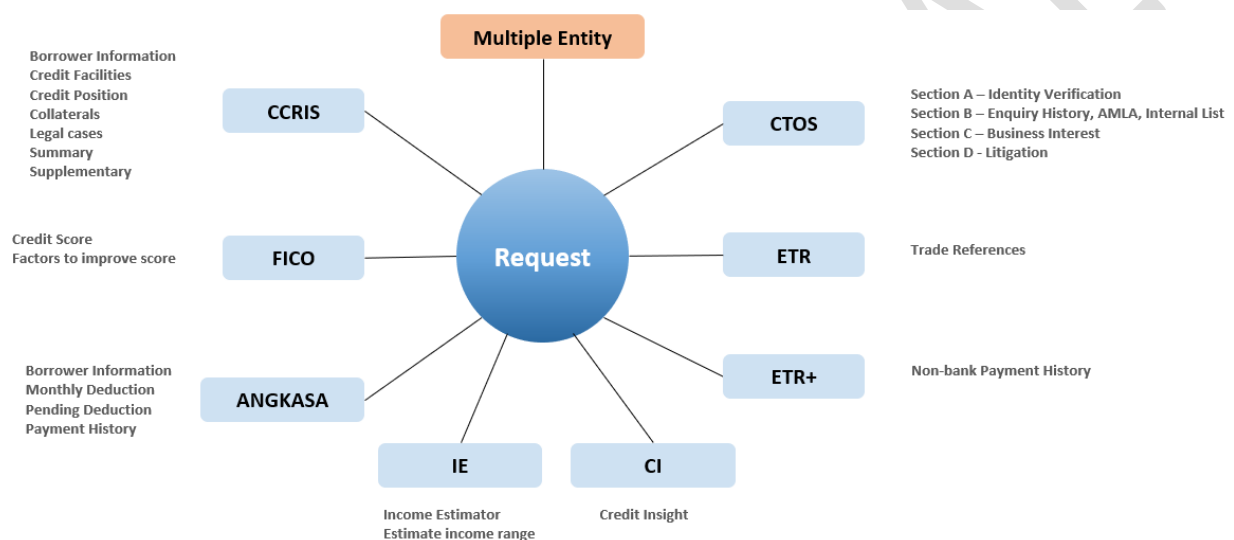
The following sub-section will explain the data which will be returned for each method name.

3.1 Overview of method 'request'

The method name 'request' will return data from eight (8) sources depending on the request submission (CTOS, AMR, CCRIS, ANGKASA, Income Estimator (IE), Credit Insight (CI), FICO and AMR Plus). This will be explained under the XML Request section.

There is a conditional result if the subject has multiple entities, all eight (8) data sources will not be returned. Instead, a list of multiple entities will be returned for SI selection. Subsequent method calls to confirm selected entity will be 'requestConfirm'

Data retrieved from this method name will require consent from the subject.



CCRIS data must be pulled together with CTOS data.

FICO data must be pulled together with CTOS data and CCRIS data.

Income Estimator (IE) and Credit Insight (CI) data must be pulled together with CCRIS and CTOS data.

3.2 Overview of method 'requestConfirm'

The method name 'requestConfirm' will only be used in the event that the requested subject has multiple entities as a result of method call 'request'. In this case, CTOS ENQWS platform returns control to the SI to select the entity.

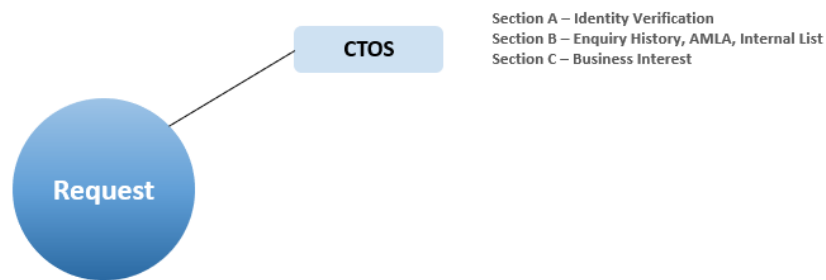
The result of this method call is the same as 'request' from the august (8) data sources.

3.3 Overview of method 'requestLite'

The method name 'requestLite' will only return data from a two (2) sources. However, the data source CTOS will be returned without Section D – Litigation.

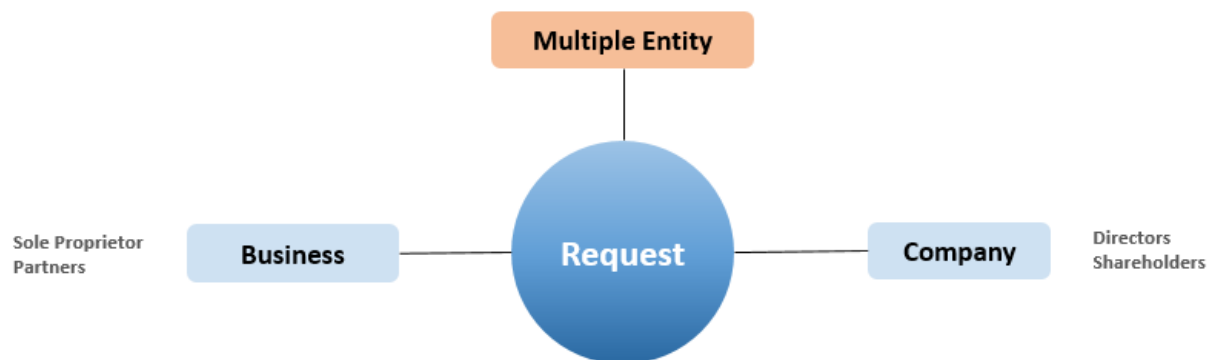
There will not be any occurrence of multiple entities as in 'request' for this method call.

Data retrieved from this method does not require consent from the subject.



3.4 Overview of method 'requestParty'

The method name 'requestParty' is an optional request. This request can be used in the event the SI requires information of the business interest parties such as directors, shareholders, sole proprietor (owner) or partner of a subject. This can be used to build the request prior to submission.



3.5 Overview of method 'requestPartyConfirm'

The method name 'requestPartyConfirm' will only be used in the event that the requested subject has multiple entities as a result of method call 'requestParty'. In this case, CTOS ENQWS platform returns control to the SI to select the entity.

The result of this method call is the same as 'requestParty'.

4 Process Flow

This section explains the flow for each method name defined in this document.

4.1 CTOS Credit Report - Individual

Subject is an individual entity, only part 'B' is applicable where a request is sent to CTOS ENQWS via method 'request'. If no multiple entities on the subject exist, the CTOS Credit Report is returned.

However, if multiple entities on the subject exist, the response will provide a list of multiple entities will be returned where the SI is required to select and resubmit request via method 'requestConfirm'.

Below are four (4) additional optional components available for this section only

1. Angkasa - You just need to turn on the include_angkasa in section 7.1.
2. Income estimator – You just need to turn on the include_ie in section 7.1 and it will automatically turn on include_ctos and include_ccris.
3. Credit Insight – You just need to turn on the include_ci in section 7.1 and it will automatically turn on include_ctos and include_ccris.
4. Trade Reference Plus – You just need to turn on the include_etr_plus in section 7.1.

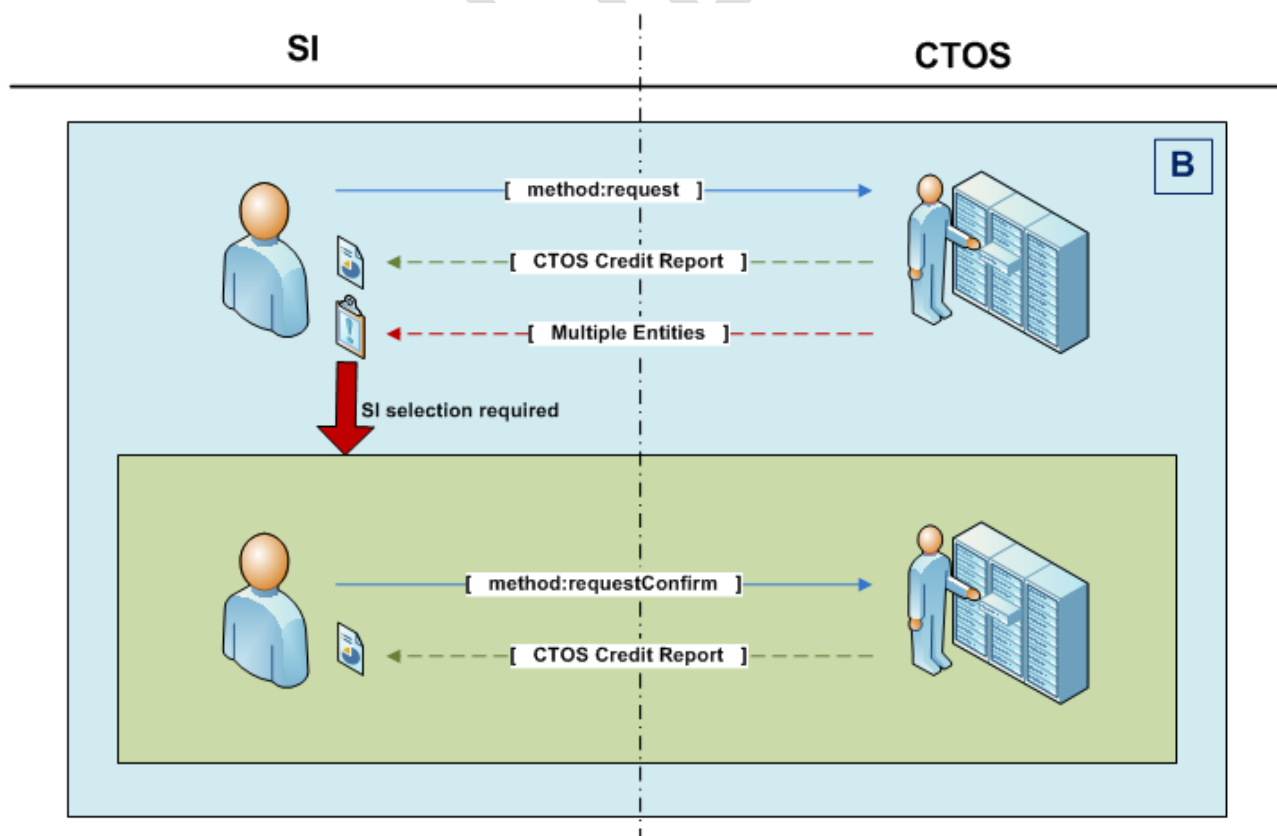
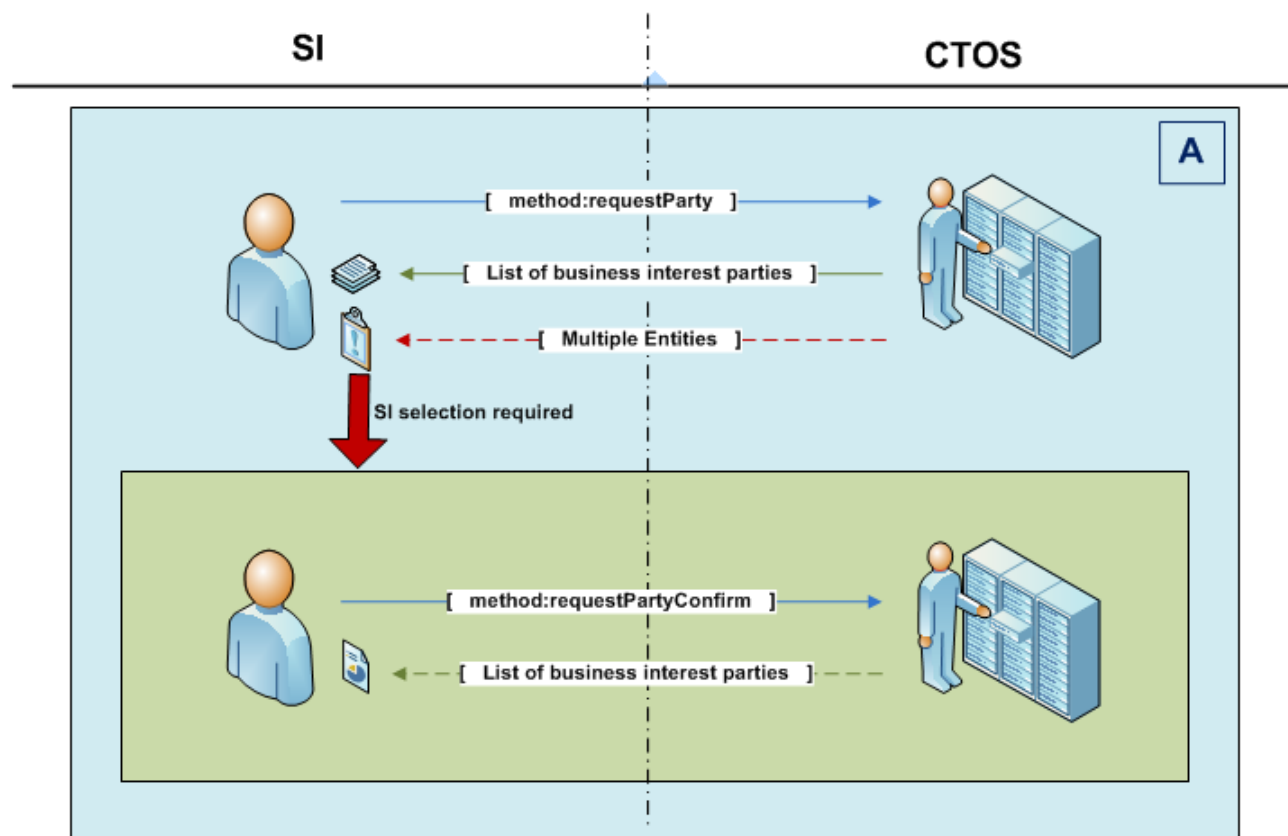
4.2 CTOS Credit Report - Company/Business/LLP

Subject is a company or business or LLP entity, both part 'A' and 'B' is applicable.

The method 'requestParty' may be used if the SI requires but does not have information of the business interest party of the subject.

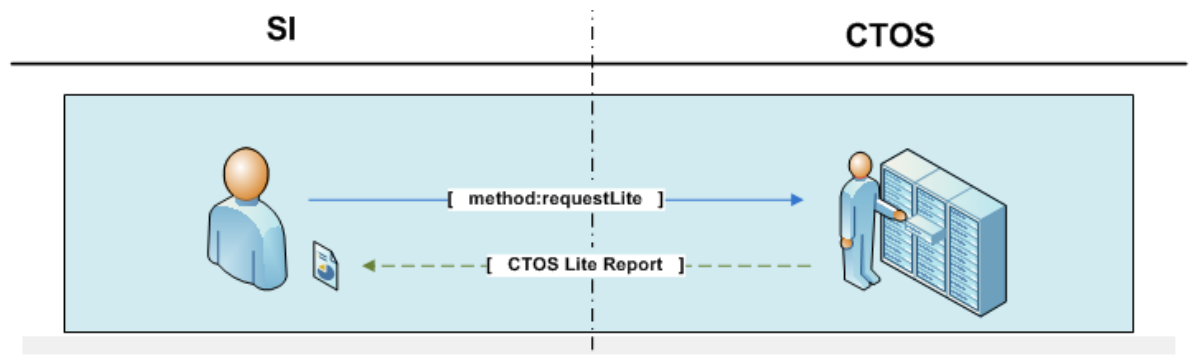
Data obtained from method 'requestParty' may then be used to create the method 'request' which will contain both company/business together with the business interest parties.

However, if multiple entities on the subject exist, the response will provide a list of multiple entities will be returned where the SI is required to select and resubmit request via method 'requestPartyConfirm'.



4.3 CTOS Lite Report

CTOS Lite report is a straight-forward request where the calls to method 'requestLite' will always return a result without multiple entities occurring.



4.4 CTOS Score Report

The process of request for CTOS Score Report will same as section 4.1 CTOS Credit Report – Individual. It just needs to turn on the include_fico in section 7.1. Include fico will always automatically turn on to include_ctos, include_trex, include_ccris, if it didn't turn on those components. The sample of request and response can be found over the sample folder.

4.5 CTOS SME Score Report for Company

The process of request for CTOS SME Score Report will same as section 4.2 CTOS Credit Report - Company/Business/LLP. You just need to turn on the include_fico in section 7.1. Include fico will always automatically turn on to include_ctos, include_trex, include_ccris, if you didn't turn on those components. The sample of request and response can be found over the sample folder.

4.6 CTOS SME Score Report for Business

The process of request for CTOS SME Score Report for business will same as section 4.2 CTOS Credit Report - Company/Business/LLP. You just need to turn on the include_fico in section 7.1. Include fico will always automatically turn on to include_ctos, include_trex, include_ccris, if you didn't turn on those components. The sample of request and response can be found over the sample folder. Below are some of the requirement need to fulfil in order to get the CTOS SME Score Report for Business entity.

1. First entity must be party type "Business" and include fico tag should set to "1"
2. Must have added at least 1 owners' entity and can be up to maximum 5 owners' entities and the owner need to be an individual.

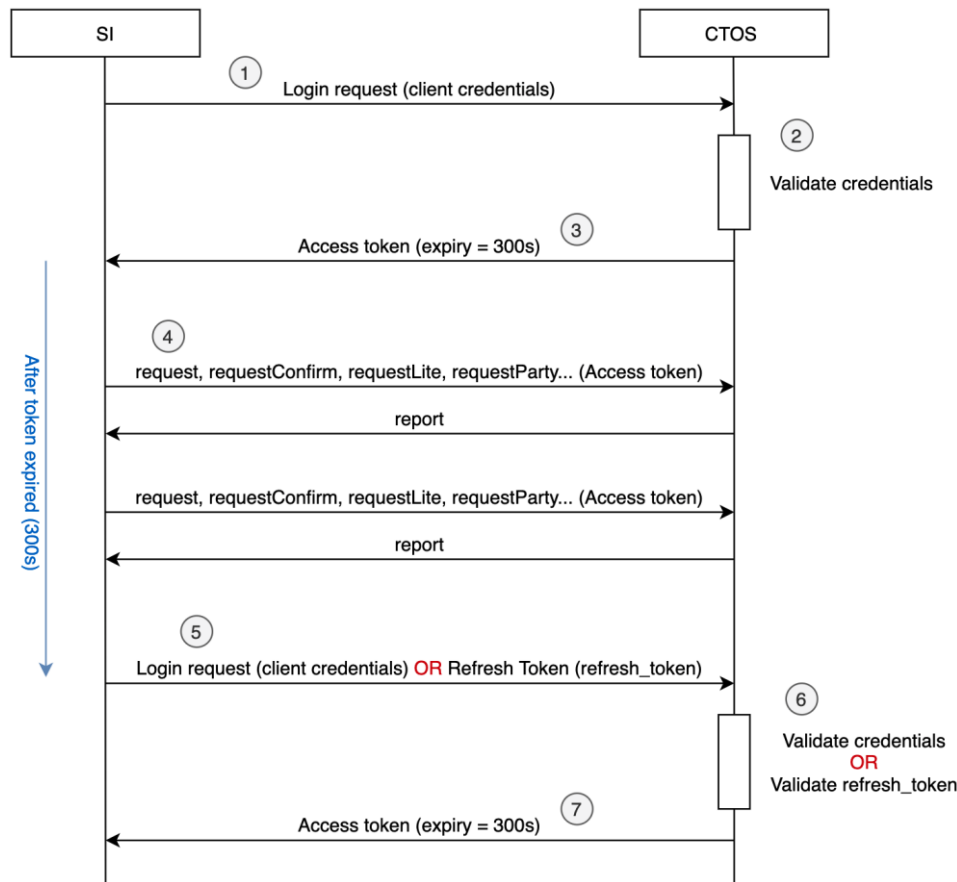
There is an additional error message to during validate the request:

- Non owners' info is added after business entity. System will throw an error message "Only individuals are allowed to be owners of a business!" (Refer to the attached sample error_response_business_fico_1.xml)
- Number of owners are exceeded 5. System will throw an error message "Maximum 5 owners allowed!" (Refer to the sample error_response_business_fico_2.xml)
- If only has 1 entity with party type business. System will throw an error message "Minimum one owner is required!" (Refer to the sample error_response_business_fico_3.xml)

5 Authentication

CTOS uses OAuth 2.0 protocol with bearer token for authentication and authorization. Calling system requires to call the CTOS authentication endpoint to fetch a short life token, and then use this token in subsequent report queries. The generation of a token requires a JWT signed token to be sent by the calling system.

The following sequence diagram shows an example of API calls with login and reports fetch:



1. SI sends its credentials to the authentication endpoint to fetch an access token. Credentials are made of:
 - client-id
 - signed JWT token (see below)
 - username
 - password

Credentials are provided upon request
2. CTOS authentication server validates the credentials provided by SI
3. If authentication is successful, a short life token is returned with an expiry duration
4. For all subsequent report enquiries, the token is passed as a Bearer authentication header
5. After token expiry time elapsed, the token cannot be used anymore and a new one must be fetched. New token can be fetched using the Login API (by sending credentials again), or by using the Refresh Token API (in which case the refresh_token is used)
6. CTOS authentication server validates the credentials (for Login API) or the refresh_token (for Refresh Token API)
7. A new token is returned, with new validity period of 300s

5.1 Private Key Generation

To ensure its confidentiality, the private key must be created by the client or SI. Openssl is a standard command line application that can generate and manipulate keys. The following commands can be used for this purpose:

```
openssl genrsa -out rsa-private.pem 2048
```

This command generates an “rsa-private.pem” file which contains the private key.

➔ This file must be kept secret and only accessible by the application.

```
openssl rsa -in rsa-private.pem -outform PEM -pubout -out rsa-public.pem
```

This command generates a “rsa-public.pem” file which contains the public key associated with the private key.

This file must be shared with CTOS to be registered to allow the authentication for that key.

5.2 Authentication to the API

Integration to the service is a 3 steps process:

1. Generate a signed JWT token using the private key file
2. Fetch a short life token using the signed JWT token
3. Fetch reports using the short life token

5.2.1 Signed JWT Token Generation

A signed JWT token contains 3 elements:

JWT Header:

The Header is the base64 representation of the string:

```
{"alg": "RS256", "typ": "JWT"}
```

JWT Payload:

The Payload is the base64 representation of the string:

```
{"jti": "${TOKEN_ID}", "sub": "${CLIENT_ID}", "iss": "${CLIENT_ID}", "aud": "https://uat-sso.ctos.com.my/auth/realms/CTOSNET/protocol/openid-connect/token", "exp": ${EXPIRY}, "iat": ${NOW}}
```

Where:

- \${TOKEN_ID} is a unique identifier for the token to be generated
- \${CLIENT_ID} is the client unique identifier (provided by CTOS)
- \${EXPIRY} is a Unix timestamp for the expiry time of the token – required to be set as current time + 300 seconds
- \${NOW} is the current timestamp in Unix format

JWT Signature:

The signature is a base64 representation of the SHA256 hash of the header.payload

Note regarding encoding:

There are 2 ways to correctly encode the JWT token and send it to the HTTP service:

- Use base64 encoding for the header, payload and signature. In this case, the token will contain characters such as +, /, = and needs to be URL-encoded prior to being sent to the HTTP header (as per standard HTTP requirement)
- Use base64url encoding for the header, payload and signature. In this case the token can be sent as is to the HTTP service as it wouldn't contain any character that requires being URL-encoded

Example script:

The following bash script shows an example how to generate a complete JWT token:

Note the `tr -- '+' '-'` command replaces characters to convert the base64 into base64url encoding:

```
#!/bin/bash
CLIENT_ID=<provided by CTOS>
PRIVATE_KEY=<path to the rsa-private.pem file>
TOKEN_EXPIRY=300

NOW=`date +%s`
EXPIRY=`expr ${NOW} + ${TOKEN_EXPIRY}`
TOKEN_ID=`echo -n tkn_${NOW}`

HEADER=`echo -n '{"alg":"RS256","typ":"JWT"}' | base64`
PAYLOAD=`echo -n
{"jti":"${TOKEN_ID}","sub":"${CLIENT_ID}","iss":"${CLIENT_ID}","aud":
"https://uat-sso.ctos.com.my/auth/realms/CTOSNET/protocol/openid-
connect/token","exp":${EXPIRY},"iat":${NOW}}' | base64`

SIGNATURE=`echo -n "${HEADER}.${PAYLOAD}" | openssl dgst -sha256 -binary -sign
${PRIVATE_KEY} | openssl enc -base64 | tr -d '\n=' | tr -- '+' '-'`

echo "${HEADER}.${PAYLOAD}.${SIGNATURE}"
```

5.2.2 Login API

The Login API call is used to get a token by using client credentials.

Request details:

- Endpoint:
 - UAT: <https://uat-sso.ctos.com.my/auth/realms/CTOSNET/protocol/openid-connect/token>
 - Production: <https://sso.ctos.com.my/auth/realms/CTOSNET/protocol/openid-connect/token>
- Parameter's format: x-www-form-urlencoded (body)
- Parameters:

Parameter name	Parameter value
grant_type	password
client_id	<i>Provided upon request</i>
username	<i>Provided upon request</i>
password	<i>Provided upon request</i>

Parameter name	Parameter value
client_assertion_type	urn:ietf:params:oauth:client-assertion-type:jwt-bearer
client_assertion	<signed JWT token from previous step>

Response details:

- HTTP Code: 200 for success, 401 for authentication error
- Response body: Token in Json format

Sample token response:

```
{
  "access_token": "eyJhbGciOiJSUzI...",
  "expires_in": 300,
  "refresh_expires_in": 1800,
  "refresh_token": "eyJhbGciOiJIUzI...",
  "token_type": "Bearer",
  "not-before-policy": 0,
  "session_state": "9d9f6030-0e56-442a-b86e-58d2e8543e10",
  "scope": "profile email"
}
```

Token elements:

Parameter name	Comments
access_token	The access token to be used as Bearer in get report requests
expires_in	Access token validity period in seconds
refresh_token	A token that can be used to refresh the access token without using username/password
refresh_expires_in	Refresh token validity period in seconds
token_type	Bearer
not-before-policy	This value is used in case there is a need to revoke existing tokens. This value is a Unix timestamp, any token provided before this timestamp becomes invalid.
session_state	This parameter can be ignored
scope	profile email – this parameter can be ignored

5.2.3 Refresh Token API

The Refresh Token API call is used to get a new access token without using client username and password.

Request details:

- Endpoint:
 - UAT: <https://uat-sso.ctos.com.my/auth/realms/CTOSNET/protocol/openid-connect/token>
 - Production: <https://sso.ctos.com.my/auth/realms/CTOSNET/protocol/openid-connect/token>
- Parameter's format: x-www-form-urlencoded (body)
- Parameters:

Parameter name	Parameter value
grant_type	refresh_token

Parameter name	Parameter value
scope	openid
client_id	<i>Same as for Login API</i>
refresh_token	<i>The "refresh_token" returned by previous Login or Refresh Token API call</i>
client_assertion_type	urn:ietf:params:oauth:client-assertion-type:jwt-bearer
client_assertion	<signed JWT token from previous step>

Response details:

- HTTP Code: 200 for success, 400 for invalid / expired token
- Response body: Token in Json format

Sample token response:

Same as "Login" API call

5.2.4 Access token usage

For each report enquiry it is required to provide a valid access_token.

The access token is sent as an HTTP header using the following format:

HTTP Header name	Header value
Authorization	Bearer <i><access_token from login API response or refresh token API response></i> Example: Bearer eyJhbGciOiJSUzI...

Note: for report fetch (request, requestConfirm, requestLite, requestParty etc...) the XML input and output formats remain unchanged.

6 Method 'requestParty' and 'requestPartyConfirm'

6.1 Request XML Format

```
<batch xmlns="[XML Namespace]">
  <company_code>[Company Code]</company_code>
  <account_no>[Account Code]</account_no>
  <user_id>[User ID]</user_id>
  <records>
    <type code="[Basic Group Code]">[Party Type]</type>
    <ic_lc>[ICLC No]</ic_lc>
    <nic_br>[NICBR No]</nic_br>
    <name>[Name]</name>
    <confirm_entity>[Entity Key Confirmation]</confirm_entity>
  </records>
</batch>
```

Variable Name	Notes	Required	Data Type	Length
XML Namespace	Default to http://ws.cmctos.com.my/ctosnet/party	Yes	String	50
Company Code	User's company code, as registered in CTOS customer database. (Provided by CTOS)	Yes	String	10
Account Code	User's account/department code. (Provided by CTOS)	Yes	String	30
User ID	User id or name. Its purpose is to allow identification of the actual person making the enquiry for traceability purpose. (Provided by CTOS)	Yes	String	50
Party Type	I – Individual C – Company B – Business L – LLP O – Others (Not I, C, B or L above)	Yes	String	1
Basic Group Code	Below is the list of basic group codes used for CCRIS entity Search 21 – Sole Proprietorship (Business) 22 – Partnership (Business) 24 - Company	Conditional	String	3
ICLC No	For individual (TYPE = I), this is the Old IC number or Police number or Army number. For company (TYPE = C), this is the local number. For business (TYPE = B), this must be blank. For LLP (TYPE = L), this must be blank. *We do not accept BNM Assigned ID	Conditional	String	12
NICBR No	For individual (TYPE = I), this is the New IC number or Passport number. For company (TYPE = C), this must be blank. For business (TYPE = B), this is the business registration number. For LLP (TYPE = L), this is the LLP registration number *We do not accept BNM Assigned ID	Conditional	Character	13
Name	Name of the subject to enquire	Yes	Character	200

Variable Name	Notes	Required	Data Type	Length
Entity Key Confirmation	Value from entity when multiple entities are detected.	Conditional (Only applicable for 'requestPartyConfirm')	Character	11

6.2 Response XML Format – Parties

Note: In XML responses, special characters are encoded to adhere to XML standards. You might encounter the & character encoded as &. This is a universal encoding practice across all XML responses.

```
<report xmlns="[XML Namespace]" version="[Version]">
  <company_code>[Company Code]</company_code>
  <account_no>[Account Code]</account_no>
  <user_id>[User ID]</user_id>
  <date>[Date request]</date>
  <time>[Time request]</time>
  <status code="[Status Code]">[Status Message]</status>
  <records>
    <type code="[Basic Group Code]">[Party Type]</type>
    <ic_lc>[ICLC No Entered]</ic_lc>
    <nic_br>[NICBR No Entered]</nic_br>
    <name>[Name Entered]</name>
  </records>
  <parties>
    <party seq="[Sequence]">
      <name>[Name]</name>
      <ic_lcno>[Old IC/Company Number]</ic_lcno>
      <nic_brno>[New IC]</nic_brno>
      <position>[Position]</position>
      <appoint>[Appoint Date]</appoint>
      <equity>[Equity]</equity>
      <party_type code="[Party Code]">[Party Desc]</party_type>
    </party>
  </parties>
</report>
```

Field	Explanation	Notes	Data Type	Length
XML Namespace	Default to http://ws.cmctos.com.my/ctosnet/party		String	50
Version	The version of XML	The version number corresponds to the version of the technical specs	String	10
Company Code		As entered in the request.	String	10
Account Code		As entered in the request.	String	30
User ID		As entered in the request.	String	50
Date request	Date of request	Format: yyyy-MM-dd	Date	
Time request	Time of request	Format: HH:mm:ss	Time	
Status Code	Status code of request	Possible values:	Integer	1

Field	Explanation	Notes	Data Type	Length
		0 = Fail 1= Success		
Status Message	Message related to status code		String	100
Basic Group Code	Basic Group Code used for CCRIS	As entered in the request.	String	3
Party Type		As entered in the request.	String	1
ICLC No Entered		As entered in the request.	String	12
NICBR No Entered		As entered in the request.	String	13
Name Entered		As entered in the request.	String	50
Sequence	Sequence number of the entities.		Integer	5
Name			String	125
Old IC/Company Number			String	20
New IC			String	20
Position			String	5
Appoint Date		Format: dd-MM-yyyy	Date	
Equity			Integer	20
Party Code			String	3
Party Desc			String	50

6.3 Response XML Format - Entities – instead of Parties

Entities are returned when the subject has multiple profiles which the system is unable to decide. The system returns control to the SI to determine which entity should be used by resubmitting request with entity key under the method requestPartyConfirm.

Note: In XML responses, special characters are encoded to adhere to XML standards. You might encounter the & character encoded as &. This is a universal encoding practice across all XML responses.

```
<report xmlns="[XML Namespace]" version="[Version]">
  <company_code>[Company Code]</company_code>
  <account_no>[Account Code]</account_no>
  <user_id>[User ID]</user_id>
  <date>[Date request]</date>
  <time>[Time request]</time>
  <status code="[Status Code]">[Status Message]</status>
  <records>
    <type code="[Basic Group Code]">[Party Type]</type>
    <ic_lc>[ICLC No Entered]</ic_lc>
    <nic_br>[NICBR No Entered]</nic_br>
    <name>[Name Entered]</name>
  </records>
  <entities>
    <subject seq="[Sequence]">
      <type>[Party Type]</type>
      <ic_lc>[ICLC No]</ic_lc>
      <nic_br>[NICBR No]</nic_br>
```

```

<name>[Name]</name>
<entity key="[Entity Key]" status="[Entity Status]" code="[Basic Group
Code]">
  <id_no1>[IDNo1]</id_no1>
  <id_no2>[IDNo2]</id_no2>
  <ent_name>[Entity Name]</ent_name>
  <industry>[Industry]</industry>
  <nationality>[Nationality]</nationality>
  <date_br>[Date BR]</date_br>
  <registered>[Registered]</registered>
  <member>[Member]</member>
  <corp_status>[Corporation Status]</corp_status>
  <industry_sector code="[Industry Sector Code]">[Industry
Sector]</industry_sector>
  <residency_status>[Residency Status]</residency_status>
</entity>
</subject>
</entities>
</report>

```

Field	Explanation	Notes	Data Type	Length
XML Namespace	Default to http://ws.cmctos.com.my/ctosnet/party		String	50
Version	The version of XML	The version number corresponds to the version of the technical specs	String	10
Company Code		As entered in the request.	String	10
Account Code		As entered in the request.	String	30
User ID		As entered in the request.	String	50
Date request	Date of request	Format: yyyy-MM-dd	Date	
Time request	Time of request	Format: HH:mm:ss	Time	
Status Code	Status code of request	Possible values: 0 = Fail 1 = Success	Integer	1
Status Message	Message related to status code		String	100
Basic Group Code	Basic Group Code used for CCRIS	As entered in the request.	String	3
Party Type		As entered in the request.	String	1
ICLC No Entered		As entered in the request.	String	12
NICBR No Entered		As entered in the request.	String	13
Name Entered		As entered in the request.	String	50

Field	Explanation	Notes	Data Type	Length
Sequence	Sequence number of the entities.		Integer	5
Party Type		As entered in the request.	String	1
ICLC No		As entered in the request.	String	12
NICBR No		As entered in the request.	String	13
Name		As entered in the request.	String	50
Entity Key	Entity Key as returned from CCRIS		String	11
Entity Status	Entity Warning as returned from CCRIS		String	20
Basic Group Code	Entity Basic Group Code as returned from CCRIS		String	3
IDNo1	As returned from CCRIS		String	20
IDNo2	As returned from CCRIS		String	20
Entity Name	As returned from CCRIS		String	125
Industry	As returned from CCRIS		String	6
Nationality	As returned from CCRIS		String	2
Date BR	As returned from CCRIS		String	10
Registered	As returned from CCRIS		String	1
Member	As returned from CCRIS		String	1
Corporation Status	As returned from CCRIS		String	1
Industry Sector Code	As returned from CCRIS		String	6
Industry Sector	As returned from CCRIS		String	120
Residency Status	As returned from CCRIS		String	1

7 Method 'request', 'requestConfirm' and 'requestLite'

7.1 Request XML Format

The input file is in a XML format as shown below:

```
<batch no="[Batch No]" output="[Output format]" xmlns="[XML Namespace]">
  <company_code>[Company Code]</company_code>
  <account_no>[Account Code]</account_no>
  <user_id>[User ID]</user_id>
  <record total>[Total Record]</record_total>
  <records>
    <type code="[Basic Group Code]">[Party Type]</type>
    <ic_lc>[ICLC No]</ic_lc>
    <nic_br>[NICBR No]</nic_br>
    <name>[Name]</name>
    <mphone_nos>
      <mphone_no>[ Mobile No]</mphone_no>
    </mphone_nos>
  </records>
</batch>
```

```

</mphone_nos>
<ref_no>[Ref No]</ref_no>
<dist_code>[Distribution Code]</dist_code>
<purpose code="[ Purpose - Code]"></purpose>
<state code="[ State - Code]"></state>
<country code="[ Country - Code]"></country>
<include_consent>[Include Consent]</include_consent>
<include_ctos>[Include CTOS]</include_ctos>
<include_trex>[ Include TREX]</include_trex>
<include_ccris sum="[Include Summary]">[Include CCRIS]</include_ccris>
<include_dcheq>[Include DCheque]</include_dcheq>
<include_ccris_supp>[Include CCRIS Supplementary]</include_ccris_supp>
<include_fico>[Include Fico]</include_fico>
<include_ssm>[Include Latest SSM]</include_ssm>
<include_ssm_retention>[Include SSM Retention]</include_ssm_retention>
<include_angkasa>[Include Angkasa]</include_angkasa>
<include_ie>[Include Income Estimator]</include_ie>
<include_ci>[Include Credit Insight]</include_ci>
<include_etr_plus>[Include ETR Plus]</include_etr_plus>
<include_analytical_product>[Include Analytical
Product]</include_analytical_product>
<confirm_entity>[Entity Key Confirmation]</confirm_entity>
</records>
</batch>

```

The beginning of the input file is denoted by the <BATCH> tag and the end of the file is denoted by the </BATCH> tag.

Within the file, there are two main sections, the header section and the body section.

7.1.1 BATCH Section

The beginning of the header section is denoted by the <BATCH> tag, whilst the end of the section is denoted by the </BATCH> tag.

The header section provides information about the user who submits the batch for enquiry. The variables batch id, company code, account code, and user id have the following formats:

Variable Name	Notes	Required	Data Type	Length
Batch No	User supplied batch number. Must be unique for the same company code and account code.	Yes	String	20
Output format	0 – XML response 2 – PDF response	Yes	String	1
XML Namespace	Default to http://ws.cmctos.com.my/ctosnet/request	Yes	String	50
Company Code	User's company code, as registered in CTOS customer database. (Provided by CTOS)	Yes	String	10
Account Code	User's account/department code. (Provided by CTOS)	Yes	String	30
User ID	User id or name. Its purpose is to allow identification of the actual	Yes	String	50

Variable Name	Notes	Required	Data Type	Length
	person making the enquiry for traceability purpose. (Provided by CTOS)			
Total Record	Total number of records in the batch	Yes	Integer	5

7.1.2 RECORDS section

The beginning of the RECORDS section is denoted by the <RECORDS> tag whilst the end is denoted by the </RECORDS> tag.

The RECORDS section contains a series of records. Each record contains all the particulars of the name of a party (individual, business or company) that the user wishes to enquire about. They are used by the system to match against the various CTOS databases to produce various reports.

Each record consists of a number of fields, each of which is enclosed within a pair of start tag and end tag. The format of the enquiry input record is defined below.

7.1.2.1 Enquiry Input Record Format

Each enquiry input record is enclosed within the <RECORDS> and </RECORDS> tags. The <RECORDS> tag may have multiple instances to support a batch enquiry.

Each record consists of the particulars of person/business/company that will be used to match against the records in CTOS database.

Variable Name	Notes	Required	Type	Length
Party Type	I – Individual C – Company B – Business L – LLP O – Others	Yes	String	1
Basic Group Code	Below is the basic group codes used for CCRIS entity Search 11 – Individual 20 – Other Non-Bank Financial Institution 21 – Sole Proprietorship (Business) 22 – Partnership (Business) 23 – Professional Bodies (include lawyers, doctors, accountants who are not registered with ROB) 24 – Company 26 – Limited Liability Partnership (LLP) 31 – Federal/Central Government (include embassies) 32 – State Governments 33 – Local Governments 34 – Statutory Bodies 41 – Trade Unions 42 – Co-operatives 43 – Societies/Associations 51 – International organizations 91 – Others	Conditional (Include CCRIS)	String	3

Variable Name	Notes	Required	Type	Length
ICLC No	For individual (TYPE = I), this is the Old IC number or Police number or Army number. For company (TYPE = C), this is the local number. For business (TYPE = B), this must be blank. For LLP (TYPE = L), this must be blank.	Conditional	String	12
NICBR No	For individual (TYPE = I), this is the New IC number or Passport number. For company (TYPE = C), this must be blank. For business (TYPE = B), this is the business registration number.** **Note: For East Malaysia's business, State - Code is mandatory to fill up, else system will treat it as West Malaysia's business For LLP (TYPE = L), this is the LLP registration number. For Others (TYPE = O), this is the ID or registration number for that organization. *We do not accept BNM Assigned ID, use passport instead	Conditional	String	20
Name	Name of the subject to enquire	Yes	String	200
Mobile No	Mobile phone number of subject, used for consent verification. CTOS will determine whether the source of the consent requires mobile number for consent verification.	Conditional (Multiple)	String	15
Ref No	User's own reference or remark This is mandatory when [Include Consent] is enabled and has value 1.	Conditional	String	50
Distribution Code	User's own distribution code e.g. branch, department	No	String	50
Purpose - Code	The purpose code of the enquiry Not applicable for method "requestLite"	Yes	String	3

Variable Name	Notes	Required	Type	Length
State - Code	Applicable if include_ccris = 1 Applicable if subject is East Malaysia's company or business	Conditional	String	5
Country - Code	The country code of the subject if "include_ccris" = 1. Not applicable for method "requestLite"	Conditional	String	5
Include Consent	This is only available if configured by CTOS where consent from subject to CTOS is provided by the subscriber together with the request. Possible values: 0 = No consent included 1 = Consent included Not applicable for method "requestLite"	Optional	String	1
Include CTOS	Indicator for response to include CTOS Report Possible values: 0 = Exclude 1 = Include	Yes. Mandatory to include if CCRIS and FICO included.	Integer	1
Include TREX	Indicator for response to include TREX results Possible values: 0 = Exclude 1 = Include	Yes	Integer	1
Include CCRIS	Indicator for response to include CCRIS Details results Possible values: 0 = Exclude 1 = Include Not applicable for method "requestLite"	Yes. Mandatory to include if FICO included.	Integer	1
Include Summary	Indicator for response to include CCRIS Summary results Possible values: 0 = Exclude 1 = Include Not applicable for method "requestLite"	No	Integer	1
Include DCheque	Indicator for response to include DCHEQS results Possible values: 0 = Exclude 1 = Include Not applicable for method "requestLite" <i>This service is decommissioned.</i>	No	Integer	1
Include CCRIS Supplementary	Indicator for response to include CCRIS Supplementary results Possible values: 0 = Exclude	No	Integer	1

Variable Name	Notes	Required	Type	Length
	1 = Include Not applicable for method "requestLite"			
Include Fico	Indicator for response to include FICO consumer & SME score Possible values: 0 = Exclude 1 = Include Not applicable for method "requestLite"	No	Integer	1
Include Latest SSM	Indicator for response to include latest SSM Info. 0 = Existing SSM Info 1 = Latest SSM Info Not applicable for East Malaysia's company and business	Conditional (CTOS defaults to '0' if not specified)	Integer	1
Include SSM Retention	Indicator for response to include retention period to pull an SSM report from achieve. Possible value: 0 - 999	Conditional (CTOS defaults to '0' if not specified)	Integer	3
Include Angkasa	Indicator for response to include Angkasa information Possible values: 0 = Exclude 1 = Include Not applicable for method "requestLite"	No	Integer	1
Include Income Estimator	Indicator for response to include Income range info. Possible value: 0 = Exclude 1 = Include Not applicable for method "requestLite"	No	Integer	1
Include Credit Insight	Indicator for response to include credit insight info. Possible value: 0 = Exclude 1 = Include Not applicable for method "requestLite"	No	Integer	1
Include ETR Plus	Indicator for response to include etr plus info. Possible value: 0 = Exclude 1 = Include Not applicable for method "requestLite"	No	Integer	1
Include Analytical Product	Value for analytical product's code. It is a repeating xml element . For example: - <include_analytical_product>TS </include_analytical_product>	No	String	15

Variable Name	Notes	Required	Type	Length
	<include_analytical_product>EWI </include_analytical_product> Not applicable for method "requestLite"			
Entity Key Confirmation	Value from entities when multiple entities is detected. Not applicable for method "requestLite"	Conditional (Only applicable for 'requestConfirm')	String	11

7.2 Response XML Format

For each request submitted, only one response output will be generated. Response will be returned as **Base 64 Binary** as this is to cater for the multiple output formats.

Note: In XML responses, special characters are encoded to adhere to XML standards. You might encounter the & character encoded as &. This is a universal encoding practice across all XML responses.

The XML Output file format is shown below:

```
<report version="[Version]"
  xmlns="http://ws.cmctos.com.my/ctosnet/response">
  <enq_report id="[Report Unique ID]" report_type="[Report Type]"
    title="[Title]">
    <header>
      <user id="[User ID]">[User Name]</user>
      <company id="[Company Code]">[Company Name]</company>
      <account>[Account Code]</account>
      <tel>[Company Telephone]</tel>
      <fax>[Company Fax]</fax>
      <enq_date>[Report Date]</enq_date>
      <enq_time>[Report Time]</enq_time>
      <enq_status code="[Status Code]">[Status Message]</enq_status>
    </header>
    <summary>
      <enq_sum ptype="[Party Type]" pcode="[Basic Group Code]"
        seq="[Record Sequence]">
        <name>[Subject Name]</name>
        <ic_lcno>[Subject IC/LC]</ic_lcno>
        <nic_brho>[Subject NIC/BR]</nic_brho>
        <stat>[Summary Status]</stat>
        <dd_index>[Due Diligence Index]</dd_index>
        <fico_index score="[FICO Score]">
          <fico_factor code="[FICO Factor Code]">[FICO Factor]</fico_factor>
        </fico_index>
      </enq_sum>
    </summary>
  </enq_report>
</report>
```

```

    <income_estimator>
      <Lower_Band>[Lower Band]</Lower_Band>
      <Upper_Band>[Upper Band]</Upper_Band>
      <IE_Status code="[IE Status Code]">[IE Status
Description]</IE_Status>
    </income_estimator>
  <ci>
    <ci1_all_facility_24m_statok>[CI1 All Facility 24m
Statok]</ci1_all_facility_24m_statok>/>
    <ci2_cc_facility_3m_statok>[CI2 CC Facility 3m
Statok]</ci2_cc_facility_3m_statok>/>
    <ci3>[CI3]</ci3>
    <ci4>[CI4]</ci4>
    <ci5>[CI5]</ci5>
    <ci6>[CI6]</ci6>
    <ci7>[CI7]</ci7>
    <ci8>[CI8]</ci8>
    <ci9>[CI9]</ci9>
    <ci10>[CI10]</ci10>
    <ci11>[CI11]</ci11>
    <ci12>[CI12]</ci12>
    <ci13>[CI13]</ci13>
    <ci14>[CI14]</ci14>
    <ci15>[CI15]</ci15>
    <ci16>[CI16]</ci16>
    <ci17>[CI17]</ci17>
    <ci18>[CI18]</ci18>
    <ci19>[CI19]</ci19>
    <ci20>[CI20]</ci20>
    <ci21>[CI21]</ci21>
    <ci22>[CI22]</ci22>
    <ci23>[CI23]</ci23>
    <ci24>[CI24]</ci24>
    <ci25>[CI25]</ci25>
    <ci26>[CI26]</ci26>
    <ci27>[CI27]</ci27>
    <ci28>[CI28]</ci28>
    <ci29>[CI29]</ci29>
    <ci30>[CI30]</ci30>
    <ci31>[CI31]</ci31>
    <ci32>[CI32]</ci32>
    <ci33>[CI33]</ci33>
    <ci34>[CI34]</ci34>
    <ci35>[CI35]</ci35>
    <ci36>[CI36]</ci36>
  </ci>
  <mphone_nos>
    <mphone_no>[Mobile Number]</mphone_no>
  </mphone_nos>
  <ref_no>[Reference Number]</ref_no>
  <dist_code>[Distribution Code]</dist_code>
  <purpose code="[Purpose Code]">[Purpose Desc] </purpose>
  <state code="[State Code]">[State Desc] </state>
  <country code="[ Country Code]">[Country Desc]</ country >
  <include_consent>[Include Consent]</include_consent>
  <include_ctos>[Include CTOS]</include_ctos>
  <include_trex>[Include TREX]</include_trex>
  <include_ccris sum="[Include Summary]">[Include CCRIS]
</include_ccris>
  <include_dcheq>[Include DCheque]</include_dcheq>
  <include_ccris_supp>
    [Include CCRIS Supplementary]
  </include_ccris_supp>

```



```

        </include_ccris_supp>
        <include_fico>[Include Fico]</include_fico>
        <include_ssm>[Include Latest SSM]</include_ssm>
        <include_ssm_retention>[Include SSM
Retention]</include_ssm_retention>
        <include_angkasa>[Include Angkasa]</include_angkasa>
        <include_ie>[Include Income Estimator]</include_ie>
        <include_ci>[Include Credit Insight]</include_ci>
        <include_etr_plus>[Include ETR Plus]</include_etr_plus>
        <include_analytical_product>[Include Analytical
Product]</include_analytical_product>
        <confirm_entity>[Entity Key Confirmation]</confirm_entity>
        <mdi_status code="[MDI code]">[MDI Message]</mdi_status>
        <enq_status code="[Status Code]">[Status Message]</enq_status>
        <enq_code code="[Report Code]">[Report Description]</enq_code>
        <analytical_products>
            <analytical_product>
                <type version="[Type Version]" code="[Type Code]">[Type
Name]</type>
                <score>[Type Score]</score>
                <score_reason_code code="[Type Score Reason Code]">[Type Score
Reason Description]</score_reason_code>
                <exclusion_code code="[Type Score Exclusion Code]">[Type Score
Exclusion Description]</exclusion_code>
            </analytical_product>
        </analytical_products>
    </enq_sum>
</summary>
<enquiry seq="[Record Sequence]">
    <section_summary/>
    <section_a data="[Availability]" title="[Title]" />
    <section_b data="[Availability]" title="[Title]" />
    <section_c data="[Availability]" title="[Title]" />
    <section_d data="[Availability]" title="[Title]" />
    <section_d2 data="[Availability]" title="[Title]" />
    <section_d4 data="[Availability]" title="[Title]" />
    <section_ccris data="[Availability]" title="[Title]" />
    <section_dcheqs data="[Availability]" title="[Title]" />
    <section_ccris_supp data="[Availability]" title="[Title]" />
    <section_angkasa data="[Availability]" title="[Title]" />
    <section_etr_plus data="[Availability]" title="[Title]" />
    <section_e data="[Availability]" title="[Title]" />
</enquiry>
<entities>
    <subject seq="[Sequence]">
        <type>[Party Type]</type>
        <ic_lc>[ICLC No]</ic_lc>
        <nic_br>[NICBR No]</nic_br>
        <name>[Name]</name>
        <entity key="[Entity Key]" status="[Entity Status]"
            code="[Basic Group Code]">
            <id_no1>[IDNo1]</id_no1>
            <id_no2>[IDNo2]</id_no2>
            <ent_name>[Entity Name]</ent_name>
            <industry>[Industry]</industry>
            <nationality>[Nationality]</nationality>
            <date_br>[Date BR]</date_br>
            <registered>[Registered]</registered>
            <member>[Member]</member>
            <corp_status>[Corporation Status]</corp_status>
            <industry_sector code="[Industry Sector Code]">

```

```

        [Industry Sector]
      </industry_sector>
      <residency_status>[Residency Status]</residency_status>
    </entity>
  </subject>
</entities>
</enq_report>
</report>

```

7.2.1 ENQUIRY Report

The <enq_report> tag denotes the start of a new CTOS Report, whilst the </enq_report> tag denotes the end of the CTOS Report. The <enq_report> section can be repeated multiple times which denotes multiple records.

Field	Explanation	Notes
Version	The version of the XML	The version number corresponds to the version of the technical specs.
Report Unique ID	Unique ID for this report	This is a key generated by CTOS to uniquely identify the report.
Title	Title of this report	
Report Type	Type of Report	Currently, there are two types of report. report_type = "CTOS" (CTOS main report) report_type = "ETR" (Trade Reference Report).

7.2.2 HEADER Section

The **header** section contains the details of the enquirer, i.e. the CTOS user who make the enquiry. The **header** section should only appear once in the entire XML file.

Field	Explanation	Notes	Data Type	Length
User ID	CTOS user's (the enquirer's) ID as in CTOS customer database	Provided by CTOS	String	12
User name	CTOS user's (the enquirer's) name as in CTOS customer database	Provided by CTOS	String	50
Company Code	CTOS user's (the enquirer's) company ID as in CTOS customer database	Provided by CTOS	String	10
Company Name	CTOS user's (the enquirer's) company name as in CTOS customer database		String	255
Account Code	CTOS user's (the enquirer's) account ID as in CTOS customer database	Provided by CTOS	String	15

Field	Explanation	Notes	Data Type	Length
Company Telephone	CTOS user's (the enquirer's) telephone number as in CTOS customer database		String	40
Company Fax	CTOS user's (the enquirer's) fax number as in CTOS customer database		String	20
Report Time	The time the enquiry was done	Format = hh:mm:ss	Time	
Report Date	The date the enquiry was done	Format = yyyy-MM-dd	Date	
Status Code	Overall status code of the enquiry	0 – Fail 1 – Success	Integer	1
Status Message	Overall status message of the Enquiry	In the event Status Code (Fail), a message will display the reason of failure	String	200

User should always check the value of the returned status code to find out whether the enquiry is successful. Status Message provides some descriptive explanation of the error condition.

User should be aware that there is another possible source of error. The web server may encounter some kinds of errors and returns a HTML error page immediately. Example is Unauthorized (401), Internal Server Error (500).

Any software agent that tries to automate the process should take this into consideration and handle the condition accordingly.

7.2.3 SUMMARY Section

The **summary** section contains the summary information of the enquiries made by the user. It should appear only once in the entire XML file.

The summary section is made up of one or more of the **enq_sum** record. Each **enq_sum** corresponds to one enquiry made by the user. It has the following attributes:

Field	Explanation	Notes	Data Type	Length
Record Sequence	Sequence number of the enquiry.		Integer	5
Party Type	Entity type of subject enquired.	Possible values: I – Individual C – Company B – Business L – LLP O – Others	String	1
Basic Group Code	Basic Group Code used for CCRIS	As entered in the request	String	3

Field	Explanation	Notes	Data Type	Length
Subject Name	The name of the subject enquired.	As entered in the request.	String	100
Subject IC/LC	For individual (Party Type = I), this is the Old IC number or Police number or Army number. For company (Party Type = C), this is the local number. For business (Party Type = B), this must be blank. For LLP (Party Type = L), this must be blank.	As entered in the request.	String	12
Subject NIC/BR	For individual (Party Type = I), this is the New IC number or Passport number. For company (Party Type = C), this must be blank. For business (Party Type = B), this is the business registration number. ** **Note: For East Malaysia's business, State - Code is mandatory to fill up, else system will treat it as West Malaysia's business For LLP (Party Type = L), this is the LLP registration number.	As entered in the request.	String	20
Summary Status	The status of the enquiry result.	Refer to " 11. Summary Section Status Codes ".	String	4
Due Diligent Index	This is a CTOS index of the severity of the record.	Refer to " 13. Due Diligent Index ".	String	4
FICO Score	Three digit FICO score		String	3
FICO Factor Code	Code of FICO Factor		String	2
FICO Factor	FICO factor description for score improvement	Multiple values	String	200
Lower Band	Minimum income range		String	10
Upper Band	Maximum income range		String	10
IE Status Code	Code of IE status		String	2
IE Status Description	Description of IE status		String	100
CI1 All Facility 24m Statok	Credit Insight Variable for all facilities with 24 months		String	10
CI2 CC Facility 3m Statok	Credit Insight Variable for credit card in the recent 3 months		String	10

Field	Explanation	Notes	Data Type	Length
CI3	Credit Insight variable for pawn broking in the last 24 months		String	10
CI4	Credit Insight variable for secured loan – Housing Loan in the last 24 months		String	10
CI5	Credit Insight variable for secured loan – Other than Housing Loan in the last 24 months		String	10
CI6	Credit Insight variable for Historical CCRIS Presence of Unsecured Term Facility and Conduct of Account Last 12 Months		Integer	10
CI7	Credit Insight variable for Historical CCRIS Utilization Trend of Revolving Facilities Last 12 Months		Integer	10
CI8	Credit Insight variable for Number of unsecured products with status O, K in last 3 months		String	10
CI9	Credit Insight variable for Number of months with a decrease in Credit Card balance in last 24 months		String	10
CI10	FLAG_SETTLE_L15m : Any facility that has been settled in last 15 months		String	10
CI11	FLAG_IAGE1_P12M : Flag of customer hit IIA minimum 1 for any facility		String	10
CI12	CC_UTIL_OPEN : Opened Credit Card Utilisation		String	10
CI13	OLDEST_OPEN_MOB: Oldest Months available across any facility		String	10
CI14	MAX_MG_MOB : Oldest Mortgage Months available		String	10
CI15	Score_AI_V1: This is a first version of tier which consist of a matrix CTOS score vs AI Note: If CI15 is opted, score and AI products will be auto enabled		String	10
CI16	Reserved		String	10
CI17	Reserved		String	10
CI18	Reserved		String	10

Field	Explanation	Notes	Data Type	Length
CI19	Reserved		String	10
CI20	Reserved		String	10
CI21	Reserved		String	10
CI22	Reserved		String	10
CI23	Reserved		String	10
CI24	Reserved		String	10
CI25	Reserved		String	10
CI26	Reserved		String	10
CI27	Reserved		String	10
CI28	Reserved		String	10
CI29	Reserved		String	10
CI30	Reserved		String	10
CI31	Reserved		String	10
CI32	Reserved		String	10
CI33	Reserved		String	10
CI34	Reserved		String	10
CI35	Reserved		String	10
CI36	Reserved		String	10
Mobile Number		As entered in the request.	String	30
Reference Number		As entered in the request.	String	50
Distribution Code		As entered in the request.	String	50
Purpose Code	Not applicable for method "requestLite"	As entered in the request.	String	3
Purpose Desc	Not applicable for method "requestLite"	Returned by CTOS as specified in the Purpose Code	String	120
State Code	Applicable if include_ccris = 1 Applicable if subject is East Malaysia's company or business	As entered in the request.	String	3
State Desc	Applicable if include_ccris = 1 Applicable if subject is East Malaysia's company or business	Returned by CTOS as specified in the State Code	String	50
Country Code	Applicable if include_ccris = 1	As entered in the request.	String	3
Country Desc	Applicable if include_ccris = 1	Returned by CTOS as specified in the Country Code	String	50
Include Consent	Not applicable for method "requestLite"	As entered in the request.	Integer	1

Field	Explanation	Notes	Data Type	Length
Include CTOS		As entered in the request.	Integer	1
Include TREX		As entered in the request.	Integer	1
Include CCRIS	Not applicable for method "requestLite"	As entered in the request.	Integer	1
Include Summary	Not applicable for method "requestLite"	As entered in the request.	Integer	1
Include Dcheque	Not applicable for method "requestLite" This service is decommissioned.	As entered in the request.	Integer	1
Include CCRIS Supplementary	Not applicable for method "requestLite"	As entered in the request.	Integer	1
Include FICO	Not applicable for method "requestLite"	As entered in the request.	Integer	1
Include Latest SSM	Not applicable for East Malaysia's company and business	Value might change depend on condition on last updated date on ssm data and setting inside the include_ssm_retention	Integer	1
Include SSM Retention		As entered in the request.	Integer	3
Include Angkasa	Not applicable for method "requestLite"	As entered in the request.	Integer	1
Include Income Estimator	Not applicable for method "requestLite"	As entered in the request.	Integer	1
Include Credit Insight	Not applicable for method "requestLite"	As entered in the request.	Integer	1
Include ETR Plus	Not applicable for method "requestLite"	As entered in the request. OR Return 2 if user account is not an ETR Plus contributor.	Integer	1
Include Analytical Product	Value for analytical product's code. It is a repeating xml element . For example: - <include_analytical_product>TS </include_analytical_product> <include_analytical_product>EWI </include_analytical_product> Not applicable for method "requestLite"	No	String	15
Entity Key Confirmation	Not applicable for method "requestLite"	As entered in the request. OR	String	11

Field	Explanation	Notes	Data Type	Length
		As per returned single entity		
MDI Code	Returns the status of the CTOS Lite status Applicable only for CTOS Lite report.	Possible values: 0: Complete - no litigation record and no etr records. 1: Fresh MDI Search Recommended - have litigation records (only bankruptcy and wind up records) 2: Furtherdue diligence with consent is recommended - have etr records or litigation records (except bankruptcy and wind up records)	Integer	1
MDI Message	Returns the message based on the MDI Code. Applicable only for CTOS Lite report.		String	100
Status Code	Returns the status of the record search.	Possible values: 0: Failed 1: Success 2: Multiple Entities	Integer	1
Status Message	Returns the message based on the Status Code.		String	110
Report Code	Returns the report type	Possible values: 1: CTOS Report 2: CTOS Lite Report 3: CTOS CCRIS Report 4: CTOS FICO Report 11: CTOS SME FICO Report	Integer	2
Report Description	Description of the report based on the Report Code		String	100
Type Code	Analytical product code		String	15
Type Name	Analytical product name		String	100
Type Version	Analytical product's version		String	3
Type Score	Analytical product score value		String	3
Type Score Reason Code	Analytical product score reason code It is a repeating xml element . For example: - <score_reason_code code="D1">Not enough information...		String	2

Field	Explanation	Notes	Data Type	Length
	<code></score_reason_code></code> <code><score_reason_code</code> <code>code="E1">Missing require info...</code> <code></score_reason_code></code>			
Type Score Reason Description	Analytical product score reason description It is a repeating xml element . For example: - <code><score_reason_code</code> <code>code="D1">Not enough information...</code> <code></score_reason_code></code> <code><score_reason_code</code> <code>code="E1">Missing require info...</code> <code></score_reason_code></code>		String	200
Type Score Exclusion Code	Analytical product score exclusion code		String	2
Type Score Exclusion Description	Analytical product score exclusion description		String	200

7.2.4 ENQUIRY Section

The `<enq_report>` will only be available if the request XML specified [Include CTOS] as 1. This is to indicate that the response will include data from CTOS Data Systems Sdn Bhd (CRA).

The **enquiry** section can appear as many times as required in the XML file. Each **enquiry** section corresponds to one name enquired. It has the following attributes:

Field	Explanation	Notes	Data Type	Length
Record Sequence	Sequence number of the enquiry.	The enq_seq corresponds to the seq_no in the summary section.	Integer	5

Each **enquiry** section is made up of subsections called **section_x** where **x** refers to A, B, C, D, D2, D4, CCRIS, Dcheqs, CCRIS Supp, Angkasa, ETRPLUS and E. The **section_x** has the following attributes:

Field	Explanation	Notes	Data Type	Length
Availability	Flag to determine whether there was a hit on this section.	false = No data available true = Data available	Boolean	
Title	Title of this section.		String	200

7.2.5 ENTITIES Section

Field	Explanation	Notes	Data Type	Length
Sequence	Sequence number of the entities.		Integer	5
Party Type		As entered in the request.	String	1

Field	Explanation	Notes	Data Type	Length
ICLC No		As entered in the request.	String	12
NICBR No		As entered in the request.	String	15
Name		As entered in the request.	String	150
Entity Key	Entity Key as returned from CCRIS		String	11
Entity Status	Entity Warning as returned from CCRIS		String	20
Basic Group Code	Basic Group Code returned from CCRIS		String	3
IDNo1	As returned from CCRIS		String	20
IDNo2	As returned from CCRIS		String	20
Entity Name	As returned from CCRIS		String	150
Industry	As returned from CCRIS		String	6
Nationality	As returned from CCRIS		String	2
Date BR	As returned from CCRIS		String	10
Registered	As returned from CCRIS		String	1
Member	As returned from CCRIS		String	1
Corporation Status	As returned from CCRIS		String	1
Industry Sector Code	As returned from CCRIS		String	6
Industry Sector	As returned from CCRIS		String	120
Residency Status	As returned from CCRIS		String	1

8 Definition of Sections

Across all defined sections, XML encoding standards are maintained. This means special characters, such as &, will appear encoded as &. This is a consistent behaviour across all XML data sections.

8.1 Section Summary

Section Summary contains a summary of legal cases, banking payment history and dishonoured cheques.

```
<section_summary>
  <ctos>
    <bankruptcy status="[Bankrupt Status]"/>
    <legal total="[Total Legal]" value="[Total Legal Amount]"/>
    <legal_personal_capacity
      total="[Total Legal Personal]"
      value="[Total Legal Personal Amount]"/>
    <legal_non_personal_capacity
      total="[Total Legal Non Personal]"
      value="[Total Legal Non Personal Amount]"/>
    <related_parties>
      <bankruptcy status="[Bankrupt Status]"/>
      <legal total="[Total Legal]" value="[Total Legal Amount]"/>
      <legal_personal_capacity
        total="[Total Legal Personal]"
        value="[Total Legal Personal Amount]"/>
      <legal_non_personal_capacity
        total="[Total Legal Non Personal]"
        value="[Total Legal Non Personal Amount]"/>
    </related_parties>
  </ctos>
  <tr>
    <trex_ref negative="[ETR - Negative]" positive="[ETR - Positive]"/>
  </tr>
  <ccris>
    <application total="[Total Application]" approved="[Approved]"
      pending="[Pending]"/>
    <facility total="[Total Facility]" value="[Total Value]"
      arrears="[Arrears]"/>
    <special_attention accounts="[Accounts]"/>
    <related_parties>
      <application total="[Total Application]"
        approved="[Approved]" pending="[Pending]"/>
      <facility total="[Total Facility]" value="[Total Value]"
        arrears="[Arrears]"/>
      <special_attention accounts="[Accounts]"/>
    </related_parties>
  </ccris>
  <dcheqs entity="[Cheques]" related_parties="[Cheques - Related Parties]"/>
</section_summary>
```

Field	Explanation	Notes	Data Type	Length
Bankrupt Status	Any record hitting bankruptcy status	Possible value: 0 – Clean 1 – Bankrupt	Integer	1
Total Legal	Total number of legal cases		Integer	5
Total legal Amount	Total value of all legal cases		Decimal	15,2
Total Legal Personal	Total number of legal cases of personal		Integer	5
Total legal Personal Amount	Total value of all legal cases of personal		Decimal	15,2
Total Legal Non Personal	Total number of legal cases of non personal		Integer	5
Total legal Non Personal Amount	Total value of all legal cases of non personal		Decimal	15,2

Field	Explanation	Notes	Data Type	Length
ETR – Negative	<u>Indicator that whether subject has ETR or not</u>	0 – subject don't have ETR record 1 - subject has ETR record	Integer	1
ETR - Positive	<u>Indicator that whether subject has ETR Plus or not</u> <i>"Reserved for ETR plus when applicable"</i>	0 – subject don't have ETR plus record 1 – subject has ETR plus record	Integer	1
Total Application	Total number of application applied		Integer	5
Approved	Total number of application approved		Integer	5
Pending	Total number of application pending		Integer	5
Total Facility	Total number of facilities		Integer	5
Total Value	Total value of all facilities		Decimal	15,2
Arrears	Flag whether there are any arrears	Possible value: 0 – No 1 – Yes	Integer	1
Accounts	Flag whether there are any special attention accounts	Possible value: 0 – No 1 – Yes	Integer	1
Cheques	Flag whether there are any dishonoured cheque by the subject/entity	Possible value: 0 – No 1 – Yes	Integer	1
Cheques – Related Parties	Flag whether there are any dishonoured cheque by the related parties	Possible value: 0 – No 1 – Yes	Integer	1

8.2 Section A

Section A contains data for identity verification against various sources from the CTOS Database.

8.2.1 Individual

```
<record rpttype="[Record Type]" pcode="[Basic Group Code]" seq="[Record Sequence]">
  <name match="[Match Status]">[Name]</name>
  <ic_lcno match="[Match Status]">[Old IC]</ic_lcno>
  <nic_brno match="[Match Status]">[New IC]</nic_brno>
  <addr date="[Last Updated Date]">[Address]</addr>
  <addr1>[Address]</addr1>
  <addr_breakdown>
    <addr1>[Address 1]</addr1>
    <addr2>[Address 2]</add2>
    <addr3>[Address 3]</addr3>
    <addr4>[Address 4]</addr4>
    <city>[City]</city>
    <state>[State]</state>
    <postcode>[Postcode]</postcode>
```

```
<country>[Country]</country>
<last_updated_date>[Last Updated Date]</last_updated_date>
</addr_breakdown>
<ccris_addresses>
  <ccris_address date="[Date Capture]">[CCRIS Address]</ccris_address>
</ccris_addresses>
<birth_date>[Date of birth]</birth_date>
<nationality>[Nationality]</nationality>
<source>[Source]</source>
<remark>[Remark]</remark>
</record>
```

Field	Explanation	Notes	Data Type	Length
Record Type	Returns the record type		String	3
Basic Group Code	As specified in the request for CTOS Report.		String	3
Record Sequence	Returns the record sequence		Integer	5
Match Status	Returns whether the field match with the data provided in the request.	Possible values: 0 = No Match 1 = Match	Integer	1
Name	Name of the subject		String	150
Old IC	Old IC of the subject		String	12
New IC	New IC of the subject		String	15
Address	Address of the subject in CTOS database		String	600
Last Updated Date	Date capture for address of the subject in CTOS database	Format: dd-MM-yyyy	Date	
Address 1	Address 1 of the subject in CTOS database		String	150
Address 2	Address 2 of the subject in CTOS database		String	150
Address 3	Address 3 of the subject in CTOS database		String	150
Address 4	Address 4 of the subject in CTOS database		String	150
City	City of the subject in CTOS database		String	50
State	State of the subject in CTOS database		String	50
Postcode	Postcode of the subject in CTOS database		String	8
Country	Country of the subject in CTOS database		String	50
Date Capture	Date capture for address of the subject returned by CCRIS	Format: dd-MM-yyyy	Date	

Field	Explanation	Notes	Data Type	Length
CCRIS Address	Address of the subject returned by CCRIS		String	150
Date of birth	Date of birth of the subject in CTOS database	Format: dd-MM-yyyy	Date	
Nationality	Nationality of the subject		String	100
Source	Source of the subject details in CTOS Database		String	50
Remark	Remarks by CTOS on the subject		String	250

8.2.2 Business/LLP

```

<record rpttype="[Record Type]" seq="[Record Sequence]">
  <name match="[Match Status]">[Name]</name>
  <addr date="[Last Updated Date]">[Address]</addr>
  <addr_breakdown>
    <addr1>[Address 1]</addr1>
    <addr2>[Address 2]</addr2>
    <addr3>[Address 3]</addr3>
    <addr4/>
    <city>[City]</city>
    <state>[State]</state>
    <postcode>[Postcode]</postcode>
    <country/>
    <last_updated_date>[Last Updated Date]</last_updated_date>
  </addr_breakdown>
  <register_no>[Business Number]</register_no>
  <additional_registration_no>[Additional Registration
No]</additional_registration_no>
  <reg_date>[Registered Date]</reg_date>
  <business_nature>[Nature of Business]</business_nature>
  <business_commenced>[Commenced Date]</business_commenced>
  <business_type>[Business Type]</business_type>
  <last_change_date>[Last Update Date]</last_change_date>
  <previous_cpo_date>[Previous SSM Source Date]</previous_cpo_date>
  <cpo_date>[SSM Source Date]</cpo_date>
  <expiry_date>[Expiry Date]</expiry_date>
  <status>[SSM Status]</status>
  <remark>[Remark]</remark>
  <partners>
    <partner seq="[Record Sequence]">
      <name>[Name]</name>
      <alias>[Alias]</alias>
      <ic>[Old IC]</ic>
      <nic>[ New IC]</nic>
      <address>[ Address] </address>
      <position>[ Position]</position>
      <appoint_date>[Appoint Date]</appoint_date>
      <remark>[Remark]</remark>
    </partner>
  </partners>
  <msic_ssms>
    <msic_ssm code="[MSIC Code]" priority="[MSIC Priority]">[MSIC
Description]</msic_ssm>
  </msic_ssms>
</record>

```

Field	Explanation	Notes	Data Type	Length
Record Type	Returns the record type		String	3
Record Sequence	Returns the record sequence		Integer	5
Match Status	Returns whether the field match with the data provided in the request.	Possible values: 0 = No Match 1 = Match	Integer	1
Name	Name of subject		String	150
Address	Business address of subject		String	600
Last Updated Date	Date capture for business address	Format: dd-MM-yyyy	Date	
Address 1	Address 1 of the business address		String	150
Address 2	Address 2 of the business address		String	150
Address 3	Address 3 of the business address		String	150
City	City of the business address		String	50
State	State of the business address		String	50
Postcode	Postcode of the business address		String	8
Business Number	Business registration number of subject		String	15
Additional Registration No	Business registration number of subject, it is additional to business number above		String	12
Registered Date	Date of business registration	Format: dd-MM-yyyy	Date	
Nature of Business	Nature of business of the subject		Text	
Commenced Date	Date of business commenced operation	Format: dd-MM-yyyy	Date	
Business Type	Type of business of the subject		String	200
Last Update Date		Format: dd-MM-yyyy	Date	
SSM Source Date		Format: dd-MM-yyyy	Date	
Previous SSM Source Date		Format: dd-MM-yyyy	Date	
Expiry Date	Expiry date for the business	Format: dd-MM-yyyy	Date	
SSM Status			String	100
Remark	CTOS remarks		String	250
Partner				
Record Sequence	Returns the record sequence		Integer	5
Name	Name of partner or sole proprietor		String	150
Alias	Alias name of partner or sole proprietor		String	50

Field	Explanation	Notes	Data Type	Length
Old IC	Old IC of partner or sole proprietor		String	12
New IC	New IC of partner or sole proprietor		String	15
Address	Address of partner or sole proprietor		String	255
Position	Position held by partner or sole proprietor	Possible values: PN = Partner SP = Sole Proprietor	String	5
Appoint Date	Appointment date of partner or sole proprietor	Format: dd-MM-yyyy	Date	
Remark	CTOS remarks		String	250
MSIC				
MSIC Code	Business Code of MSIC		String	10
MSIC Priority	Business Priority of MSIC	The smaller the value, the higher the priority.	Integer	1
MSIC Description	Business Description of MSIC		String	255

8.2.3 Company

```
<record rpttype=" [Record Type]" seq=" [Record Sequence]">
  <name match=" [Match Status]">[Name]</name>
  <ic_lcno>[Local Number]</ic_lcno>
  <additional_registration_no>[Additional Registration
No]</additional_registration_no>
  <lc_checksum>[Checksum]</lc_checksum>
  <addr date=" [Last Updated Date]">[Address]</addr>
  <addr1>[Business Address]</addr1>
  <addr_breakdown>
    <addr1>[Address 1]</addr1>
    <addr2>[Address 2]</addr2>
    <addr3>[Address 3]</addr3>
    <addr4/>
    <city>[City]</city>
    <state>[State]</state>
    <postcode>[Postcode]</postcode>
    <country/>
    <last_updated_date>[Last Updated Date]</last_updated_date>
  </addr_breakdown>
  <previous_cpo_date>[Previous SSM Source Date]</previous_cpo_date>
  <cpo_date>[SSM Source Date]</cpo_date>
  <register_date>[Registered Date]</register_date>
  <comp_category code=" [Code of Company Category]">[Company
Category]</comp_category>
  <comp_type code=" [Code of Company Type]">[Company Type]</comp_type>
  <status>[Company Status]</status>
  <type_of_business>[Nature of Business]</type_of_business>
  <remark>[Remark]</remark>
  <capital>[Capital]</capital>
  <paidUp>[Paid Up]</paidUp>
  <directors>
```



```

    <director seq="[Record Sequence]">
      <party_type>[Party Type]</party_type>
      <name>[Director Name]</name>
      <alias>[Alias]</alias>
      <additional_registration_no>[ Additional Registration
No]</additional_registration_no>
      <ic_lcno>[Director Old IC/Company Number]</ic_lcno>
      <nic_brno>[Director New IC]</nic_brno>
      <position>[Director Position]</position>
      <addr>[Director Address]</addr>
      <appoint>[Appoint Date]</appoint>
      <resign_date>[Resign Date]</resign_date>
      <equity>[Equity]</equity>
      <equity_percentage>[Equity Percentage]</equity_percentage>
      <remark>[Remark]</remark>
      <brn_ssm>[SSM Business Registration Number]</brn_ssm>
    </director>
  </directors>
  <secretaries>
    <secretary seq="[Record Sequence]">
      <name>[Secretary Name]</name>
      <alias>[ Secretary Alias]</alias>
      <ic_lcno>[Secretary Old IC]</ic_lcno>
      <nic_brno>[Secretary New IC]</nic_brno>
      <position>[Secretary Position]</position>
      <addr>[Secretary Address]</addr>
      <appoint>[Appoint Date]</appoint>
      <resign_date>[Resign Date]</resign_date>
    </secretary>
  </secretaries>
  <previous_names name="[Latest Previous Name]" change_date="[Name Change
Date]">
    <previous_name seq="[Record Sequence]">
      <name>[Previous Name]</name>
      <change_date>[Name Change Date]</change_date>
    </previous_name>
  </previous_names>
  <charges>
    <charge seq="[Record Sequence]">
      <name>[Chargee Name] </name>
      <charge_date>[Charge Date]</charge_date>
      <status>[Charge Status]</status>
      <charge_amount>[Amount]</charge_amount>
    </charge>
  </charges>
  <accounts>
    <account seq="[Record Sequence]">
      <conac>[Conac]</conac>
      <unqrep>[ Unqrep]</unqrep>
      <auditor_name>[ Auditor Name]</ auditor_name>
      <auditor_address>[ Auditor Address]</ auditor_address>
      <exempted>[ Exempted]</exempted>
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      <tabledt>[ Tabledt]</tabledt>
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      <bscibank>[ Bscibank]<bscibank>
      <totass>[ Totass]<totass>
      <curlib>[Curlib]<curlib>
      <bssl1td>[ Bssl1td]<bssl1td>

```

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    <bsclstd>[ Bsc1std]<bsclstd>
    <totlib>[ Totlib]<totlib>
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    <plnpat>[ Plnpat]<plnpat>
    <plminin>[ Plminin]<plminin>
    <plnetdiv>[ Plnetdiv]<plnetdiv>
    <plyear>[ Plyear]<plyear>
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    <profit_margin>[Profit Margin]<profit_margin>
    <return_on_equity>[Return of Equity]<return_on_equity>
    <currat>[ Currat]<currat>
    <workcap>[ Workcap]<workcap>
    <gear>[Gear]<gear>
    <dividen_pay_out>[Dividend Pay Out]</dividen_pay_out>
    <dividen_per_share>[Dividend Per Share]</dividen_per_share>
    <new_form>[SSM Form Format]</new_form>
  </account>
</accounts>
<capital_details>
  <auth_capital>[Authorized Capital]</auth_capital>
  <paidup_capital>[Paidup Capital]</paidup_capital>
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  <paidup_ordA_amount>[Paidup amount - Ordinary A]</paidup_ordA_amount>
  <paidup_ordB_amount>[Paidup amount - Ordinary B]</paidup_ordB_amount>
  <paidup_ord_divided>[Paidup divided - Ordinary]</paidup_ord_divided>
  <paidup_ordA_divided>[Paidup divided - Ordinary A]</paidup_ordA_divided>
  <paidup_ordB_divided>[Paidup divided - Ordinary B]</paidup_ordB_divided>
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  <paidup_preA_amount>[Paidup amount - Preference A]</paidup_preA_amount>
  <paidup_preB_amount>[Paidup amount - Preference B]</paidup_preB_amount>
  <paidup_pre_divided>[Paidup divided - Preference]</paidup_pre_divided>
  <paidup_preA_divided>[Paidup divided - Preference
A]</paidup_preA_divided>
  <paidup_preB_divided>[Paidup divided - Preference
B]</paidup_preB_divided>
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  <paidup_othB_amount>[Paidup amount - Others B]</paidup_othB_amount>
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  <paidup_othB_divided>[Paidup divided - Others B]</paidup_othB_divided>
  <paidup_oth_nominal>[Paidup nominal - Others]</paidup_oth_nominal>
</capital_details>
  <brn_ssm>[SSM Business Registration Number]</brn_ssm>
  <msic_ssms>
    <msic_ssm code="[MSIC Code]" priority="[MSIC Priority]">[MSIC
Description]</msic_ssm>
  </msic_ssms>
</record>

```

Field	Explanation	Notes	Data Type	Length
Record Type	Returns the record type		String	3
Record Sequence	Returns the record sequence		Integer	5
Match Status	Returns whether the field match with the data provided in the request.	Possible values: 0 = No Match 1 = Match	Integer	1
Name	Name of company		String	150
Local Number	Company registration number		String	12
Checksum	Company registration checksum		String	2
Additional Registration No	Company registration number, additional registration no for local number above		String	12
Address	Company registration address		String	400
Last Updated Date	Date capture for company registration address	Format: dd-MM-yyyy	Date	
Business Address	Company business address		String	400
Address 1	Address 1 of the company registration address		String	150
Address 2	Address 2 of the company registration address		String	150
Address 3	Address 3 of the company registration address		String	150
City	City of the company registration address		String	50
State	State of the company registration address		String	50
Postcode	Postcode of the company registration address		String	8
SSM Source Date	SSM source date	Format: dd-MM-yyyy	Date	
Previous SSM Source Date	Previous SSM source date	Format: dd-MM-yyyy	Date	
Registered Date	Company registration date	Format: dd-MM-yyyy	Date	
Code of Company Category	Category of company	Possible values: S = LIMITED BY SHARES B = LIMITED BY SHARE AND GUARANTEE G = LIMITED BY GUARANTEE U = UNLIMITED	String	1
Company Category	Description for category of company		String	30

Field	Explanation	Notes	Data Type	Length
Code of Company Type	Type of company	Possible values: R = PRIVATE LIMITED U = PUBLIC LIMITED	String	1
Company Type	Description for type of company		String	30
Company Status	Status of company	Possible values: EXISTING WINDING UP DISSOLVED DISSOLVED - CONVERSION TO LLP CEASED BUSINESS REMOVED STRIKING OFF	String	30
Nature of Business	Nature of business		Text	
Remark	CTOS remarks		String	250
Capital	Company capital		Integer	15
Paid Up	Company Paid Up		Integer	15
Director				
Record Sequence	Returns the record sequence		Integer	5
Party Type	Party type Possible values: "I", "C"		Integer	1
Name	Name of director or shareholder		String	100
Alias	Alias name of director		String	50
Old IC/Local Number	Old IC/Local Number of director or shareholder		String	12
New IC	New IC of director or shareholder		String	15
Position	Position held by of director or shareholder	Possible values: DO = Director Only SO = Shareholder Only DS = Director & Shareholder AD = Alternate Director AS = Alternate Director & Shareholder	String	3
Director Address	Address for director / shareholder		String	100
Appoint Date	Appointment date of director or shareholder	Format: dd-MM-yyyy	Date	
Resign Date	Resign date of director or shareholder	Format: dd-MM-yyyy	Date	
Equity	Equity owned by shareholder		Integer	15

Field	Explanation	Notes	Data Type	Length
Equity Percentage	Percentage of equity owned by shareholder		Decimal	3,2
Remark	CTOS Remarks		String	250
Secretary				
Record Sequence	Returns the record sequence		Integer	5
Secretary Name	Name of director or secretary		String	100
Secretary Alias	Alias name of secretary		String	50
Secretary Old IC	Old IC of secretary		String	12
Secretary New IC	New IC of secretary		String	15
Secretary Position	Position held by of secretary	Possible values: SC = Secretary	String	3
Secretary Address	Address for secretary		String	100
Appoint Date	Appointment date of secretary	Format: dd-MM-yyyy	Date	
Resign Date	Resign date of secretary	Format: dd-MM-yyyy	Date	
Previous Names				
Latest Previous Name	The latest changed name		String	150
Name Change Date	Date the name was changed	Format: dd-MM-yyyy	Date	
Record Sequence	Sequence of all name changes		Integer	5
Previous Name	Full list of all changes in name		String	150
Charge				
Record Sequence	Returns the record sequence		Integer	5
Chargee Name	Name of charge		String	150
Charge Date	Date of charge	Format: dd-MM-yyyy	Date	
Charge Status	Status of the charge	Possible values: U: Unsatisfied S: Fully Satisfied R: Fully Released C/A: Fully Ceased B/T: Cancellation M: Partly Satisfied Q/P: Partly Released	String	1
Amount			Decimal	15,2
Account				
Record Sequence	Returns the record sequence		Integer	5

Field	Explanation	Notes	Data Type	Length
Conac	Consolidated Account indicator	Possible values: true false	String	10
Unqrep	Unqualified Reports indicator	Possible values: true false	String	10
Auditor Name	Auditor's firm name		String	150
Auditor Address	Auditors's firm address		String	255
Exempted	Exempt Private indicator	Possible values: true false	String	10
Pldd	Financial Year End	Format: dd-MM-yyyy	Date	
Bsdd	Balance Sheet Financial Year	Format: yyyy- MM-dd	Date	
TableDt	Date of Tabling	Format: dd-MM-yyyy	Date	
Bsfatot	Fixed Assets		Decimal	15,2
Othass	Other Assets		Decimal	15,2
Bscatot	Current Assets		Decimal	15,2
Bsclbank	Non-current Assets		Decimal	15,2
Totass	Total Assets		Decimal	15,2
Curlib	Current Liability		Decimal	15,2
Bssltd	Long-term Liability		Decimal	15,2
Bsclstd	Non-current Liability		Decimal	15,2
Totlib	Total Liability		Decimal	15,2
Bsqpuc	Paid Up		Decimal	15,2
Bsqres	Share Prem/Reserves		Decimal	15,2
Bsqpro	Rev & Other Reserves		Decimal	15,2
Networth	Networth		Decimal	15,2
Bsqmint	Minority Interest		Decimal	15,2
Tot	Total		Decimal	15,2
Turnover	Turnover		Decimal	15,2
Plnpbt	Profit Before tax		Decimal	15,2
Plnpat	Profit After tax		Decimal	15,2
Plminin	Minority Interests		Decimal	15,2
Plnetdiv	Net dividend		Decimal	15,2
Plyear	Profit and Loss of the year		Decimal	15,2
Turnover Growth	Turnover Growth		Decimal	15,2
Profit Margin	Profit Margin		Decimal	15,2
Return of Equity	Return of Equity		Decimal	15,2
Currat	Current Ratio		Decimal	15,2
Workcap	Working Capital		Decimal	15,2
Gear	Gearing		Decimal	15,2

Field	Explanation	Notes	Data Type	Length
Dividend Pay Out	Dividend Pay Out		Decimal	15,2
Dividend per Share	Dividend per Share		Decimal	15,2
SSM Form Format	Format used for reporting	Possible Values: 0: Old Form 1: New Form	Integer	1
Share Capital Details				
Authorized Capital	Total authorized capital as per SSM		Integer	15
Paidup Capital	Total paidup capital as per SSM		Integer	15
Paidup amount - Ordinary	Amount of ordinary share		Decimal	15,2
Paidup amount – Ordinary A	Amount of ordinary A share		Decimal	15,2
Paidup amount – Ordinary B	Amount of ordinary B share		Decimal	15,2
Paidup divided - Ordinary	Amount of ordinary share divided into		Integer	15
Paidup divided – Ordinary A	Amount of ordinary A share divided into		Integer	15
Paidup divided – Ordinary B	Amount of ordinary B share divided into		Integer	15
Paidup nominal - Ordinary	Nominal value per ordinary share		Integer	15
Paidup amount - Preference	Amount of preference share		Decimal	15,2
Paidup amount – Preference A	Amount of preference A share		Decimal	15,2
Paidup amount – Preference B	Amount of preference B share		Decimal	15,2
Paidup divided - Preference	Amount of preference share divided into		Integer	15
Paidup divided – Preference A	Amount of preference A share divided into		Integer	15
Paidup divided – Preference B	Amount of preference B share divided into		Integer	15
Paidup nominal - Preference	Nominal value per preference share		Integer	15
Paidup amount - Others	Amount of others share		Decimal	15,2
Paidup amount – Others A	Amount of others A share		Decimal	15,2
Paidup amount – Others B	Amount of others B share		Decimal	15,2
Paidup divided - Others	Amount of others share divided into		Integer	15
Paidup divided – Others A	Amount of others A share divided into		Integer	15

Field	Explanation	Notes	Data Type	Length
Paidup divided – Others B	Amount of others B share divided into		Integer	15
Paidup nominal - Others	Nominal value per others share		Integer	15
SSM Business Registration Number	Business registration number as per SSM Format		String	12
MSIC				
MSIC Code	Business Code of MSIC		String	10
MSIC Priority	Business Priority of MSIC	The smaller the value, the higher the priority.	Integer	1
MSIC Description	Business Description of MSIC		String	255

8.3 Section B

Section B contains information from subscriber's monitoring and subscriber's internal list. It will also contain an Enquiry History.

8.3.1 Enquiry History

This section provides a history of enquiries (CTOS Report) performed on the subject by different entities. The number of enquiries that the Subject has been searched by financial institutions, non-financial institutions (corporate entities) and legal firms.

```
<history rpttype="[Record Type]" seq="[Record Sequence]" year="[Year]">
  <period month="[Month]">
    <entity type="[Entity Type]" value="[Value]" />
  </period>
</history>
```

Field	Explanation	Notes	Data Type	Length
Record Type	Returns the record type		String	3
Record Sequence	Returns the record sequence		Integer	5
Year	Year of enquiry history	Format: yyyy	Date	
Month	Month of enquiry history	Possible values: 1 to 12	Integer	2
Entity Type	Type of entity	Possible values: - FI = Banks - NONFI = SME - LAWYER	String	15
Value	Count of enquiry history hit		Integer	15

8.3.2 Individual – Monitoring and Internal List

```
<record rpttype="[Record Type]" seq="[Record Sequence]">
  <title>[Title]</title>
  <name>[Name]</name>
  <alias>[Alias]</alias>
  <ic_lcno>[Old IC Number]</ic_lcno>
  <nic_brno>[New IC Number]</nic_brno>
  <ref>[Reference]</ref>
  <firm>[Firm]</firm>
  <remark1>[Remark 1]</remark1>
  <remark2>[Remark 2]</remark2>
  <remark3>[Remark 3]</remark3>
  <amount>[Amount]</amount>
  <entry>[Date of Entry]</entry>
</record>
```

Field	Explanation	Notes	Data Type	Length
Record Type	Returns the record type		String	3
Record Sequence	Returns the record sequence		Integer	5
Title	Title of record		String	150
Name	Subject's name		String	150
Alias	Subject's alias name		String	100
Old IC Number	Subject's old identification number		String	12
New IC Number	Subject's new identification number		String	15
Reference	Subscriber internal reference		String	50
Firm			String	100
Remark 1	Subscriber's remark field 1		String	100
Remark 2	Subscriber's remark field 2		String	100
Remark 3	Subscriber's remark field 3		String	100
Amount	Amount		Decimal	15,2
Date of Entry	Date the record was entered by the subscriber	Format: yyyy-MM-dd	Date	

8.3.3 Company – Monitoring and Internal List

```
<record rpttype="[Record Type]" seq="[Record Sequence]">
  <title>[Title]</title>
  <name>[name] </name>
  <ic_lcno>[Company Local Number]</ic_lcno>
  <nic_brno>[Business Registration Number]</nic_brno>
  <ref>[Reference]</ref>
  <remark1>[Remark 1]</remark1>
  <remark2>[Remark 2]</remark2>
  <remark3>[Remark 3]</remark3>
```

```
<amount>[Amount]</amount>
<entry>[Date of Entry]</entry>
<account>[Account]</account>
</record>
```

Field	Explanation	Notes	Data Type	Length
Record Type	Returns the record type		String	3
Record Sequence	Returns the record sequence		Integer	5
Title	Title of record		String	150
Name	Subject's name		String	150
Company Local Number	Company local number (SSM)		String	12
Business Registration Number	Business registration number (SSM)		String	15
Reference	Subscriber internal reference		String	50
Remark 1	Subscriber's remark field 1		String	100
Remark 2	Subscriber's remark field 2		String	100
Remark 3	Subscriber's remark field 3		String	100
Amount	Amount		Decimal	15,2
Date of Entry	Date the record was entered	Format: yyyy-MM-dd	Date	
Account	Account of submission	CTOS Account Code	String	15

8.3.4 US Treasury and UN Security Council

For financial institutions (FI), an additional database will be included in this section, US Treasury and UN Security Council.

```
<record rpttype="[Record Type]" seq="[Record Sequence]">
  <title>[Title]</title>
  <name>[Name]</name>
  <known_as>[Other Name]</known_as>
  <nationality>[Nationality]</nationality>
  <ids>
    <id seq="[Record Sequence]">[Identification ID]</id>
  </ids>
  <birth_date>[Date of Birth]</birth_date>
  <birth_place>[Place of Birth]</birth_place>
</record>
```

Field	Explanation	Notes	Data Type	Length
Record Type	Returns the record type		String	3
Record Sequence	Returns the record sequence		Integer	5

Field	Explanation	Notes	Data Type	Length
Title	Title of record		String	150
Name	Subject's Name		String	170
Other Name	Subject's other name or known as		Text	
Nationality	Subject's nationality		String	60
Date of Birth	Subject's date of birth	Format: yyyy-MM-dd	Date	
Place of Birth	Subject's place of birth		String	200
ID				
Record Sequence	Returns the record sequence		Integer	5
Identification ID	Subject's identification ID	Multiple values possible	String	150

8.4 Section C

Section C provides information on the subject's directorship and other business interest.

```

<record rpttype="[Record Type]" seq="[Record Sequence]">
  <company_name>[Company Name]</company_name>
  <local>[Company Local Number]</local>
  <locala>[Checksum]</locala>
  <additional_registration_no>[Additional Registration
No]</additional_registration_no>
  <object>[Nature of Business]</object>
  <incdate>[Incorporate Date]</incdate>
  <lastdoc>[Last Document Filed]</lastdoc>
  <appoint>[Appointed Date]</appoint>
  <resign>[Resignation Date]</resign>
  <name>[Name]</name>
  <ic_lcno>[Old IC/Company Local Number]</ic_lcno>
  <nic_brno>[New IC]</nic_brno>
  <addr>[Address]</addr>
  <position>[Position]</position>
  <cpo_date>[SSM Source Date]</cpo_date>
  <paidup>[Paid Up]</paidup>
  <shares>[Shares]</shares>
  <total_shares_percentage>[Total Shares Percentage]</
total_shares_percentage>
  <remark>[Remark]</remark>
  <protif_after_tax>[Profit after tax]</protif_after_tax>
  <latest_financial_year>[Latest Financial Year]</latest_financial_year>
  <status>[Status]</status>
  <brn_ssm>[SSM Business Registration Number]</brn_ssm>
</record>

```

Field	Explanation	Notes	Data Type	Length
Record Type	Returns the record type		String	3
Record Sequence	Returns the record sequence		Integer	5
Company Name	Name of Company		String	150

Field	Explanation	Notes	Data Type	Length
Company Local Number	Company Registration Number		String	12
Checksum	Company local number checksum		String	1
Additional Registration No	Additional registration no to the corresponding company registration number above		String	12
Nature of Business	Nature of Business		Text	
Incorporate Date	Date of incorporation	Format: dd-MM-YYYY	Date	
Last Document Filed	Last filling at SSM	Format: dd-MM-YYYY	Date	
Appointed Date	Director/Equity owner appointment date	Format: dd-MM-YYYY	Date	
Resignation Date	Director/Equity owner Resignation Date	Format: dd-MM-YYYY	Date	
Name	Director/Equity owner's Name		String	150
Old IC/Company Local Number	Director/Equity owner's Old IC/Company local number		String	12
New IC	Director/Equity owner's New IC		String	15
Address	Director/Equity owner's address		String	255
Position	Director/Equity owner's position		String	50
SSM Source Date	Data from SSM dated	Format: dd-MM-YYYY	Date	
Paid Up	Company's paid up		Integer	15
Shares	Director/Equity owner's total shares		Integer	15
Total Shares Percentage	Total shares percentage		Decimal	3,2
Remark	Additional remarks from CTOS		String	250
Profit after tax	Company profit after tax		Decimal	15,2
Latest Financial Year	Company latest financial year	Format: dd-MM-YYYY	Date	
Status	Status of company	Possible values: • EXISTING • WINDING UP • DISSOLVED • STRIKING OFF	String	50

Field	Explanation	Notes	Data Type	Length
SSM Business Registration Number	Business registration number as per SSM Format		String	12

8.5 Section D

Section D provides information on legal cases such as summons, writs, bankruptcy proceedings, foreclosure and others where the subject is the defendant.

Not applicable for method "requestLite"

```
<record rpttype="[Record Type]" seq="[Record Sequence]" status="[Status]">
  <title>[Title]</title>
  <special_remark>[Special Remark]</special_remark>
  <name match="[Match Status]">[Name]</name>
  <alias>[Alias]</alias>
  <addr>[Address]</addr>
  <ic_lcno match="[Match Status]">[Old IC/Local Number]</ic_lcno>
  <nic_brno match="[Match Status]">[New IC/Business Registration
Number]</nic_brno>
  <cpo_date>[SSM Source Date]</cpo_date>
  <position>[Position]</position>
  <local>[Company registration number]</local>
  <case_no>[Case Number]</case_no>
  <court_detail>[Court Details]</court_detail>
  <firm>[Firm]</firm>
  <independent>[Independent Advisor]</independent>
  <property_location>[Property Location]</property_location>
  <property_description>[Property Description]</property_description>
  <respondent>[Respondent]</respondent>
  <exparte>[Exparte]</exparte>
  <plaintiff>[Plaintiff]</plaintiff>
  <petitioner>[Petitioner]</petitioner>
  <assignee>[Assignee]</assignee>
  <chargee>[Chargee]</chargee>
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  </action>
</record>
```

```

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<gazette_discharge>[Gazette Discharge]</gazette_discharge>
<register_owner>[Registered Owner]</register_owner>
<object>[Nature of Business]</object>
<hear_date>[Hearing Date]</hear_date>
<appoint_date>[Date Appointed]</appoint_date>
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<strike_off_date>[Strike Off Date]</strike_off_date>
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  [Date of Letter Section 308]
</letter_date_under_308_1>
<letter_date_under_308_2>
  [Date of Notice Section 308]
</letter_date_under_308_2>
<resigned_date>[Date Resigned]</resigned_date>
<incorporate_date>[Date Incorporated]</incorporate_date>
<return_date>[Annual Return Date]</return_date>
<assignment_date>[Date of assignment]</assignment_date>
<order_for_sale_date>[Sale Order Date]</order_for_sale_date>
<action_dated>[Action Dated]</action_dated>
<outstanding_amount>[Outstanding amount]</outstanding_amount>
<shareholding>[Shareholding]</shareholding>
<amount>[Amount]</amount>
<reserve_price>[Reserved Price]</reserve_price>
<capital>[Capital]</capital>
<paidup>[Paidup]</paidup>
<remark>[Remark]</remark>
<remark1>[Remark 1]</remark1>
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<remark4>[Remark 4]</remark4>
<administrator1>[Administrator 1]</administrator1>
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<administrator3>[Administrator 3]</administrator3>
<administrator4>[Administrator 4]</administrator4>
<lawyer>
  <name>[Lawyer Name]</name>
  <add1>[Lawyer Address 1]</add1>
  <add2>[Lawyer Address 2]</add2>
  <add3>[Lawyer Address 3]</add3>
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    <nic_brno>[Other Defendant New IC]</nic_brno>
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    <nic_brno>[Director New IC]</nic_brno>
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    <shareholdings>[Director Shares]</shareholdings>
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</directors>

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```

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    <resign_date>[Director Date Resigned]</resign_date>
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</directors>
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  <comment_date>[Comment Date]</comment_date>
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<cra_cmt>
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</withheld>
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Field	Explanation	Notes	Data Type	Length
Record Type	Returns the record type		String	3
Record Sequence	Returns the record sequence		Integer	5
Status	Returns the status of the record to indicate whether the record is released or withheld due to CRA Compliance regulation.	Possible values: 0 - Withheld 1 - Released	Integer	1
Title	Title of record type		String	150
Special Remark	Special remark by CTOS		String	100
Name	Subject Name		String	150
Match Status	Status of matching	Possible values: 0 = no match 1 = match	Integer	1
Alias	Subject name alias		String	100
Address	Address of subject		String	200
Old IC/Local Number			String	12
New IC/Business Registration Number			String	15
SSM Source Date		Format: dd-MM-yyyy	Date	
Position	Position of director/shareholder		String	30
Company Registration Number	Company registration number of related entity		String	12
Date Appointed		Format: dd-MM-yyyy	Date	
Date Resigned		Format: dd-MM-yyyy	Date	
Case Number			String	50
Court Details			String	80
Firm	Company for director/shareholder		String	150
Property Location			String	200

Field	Explanation	Notes	Data Type	Length
Property Description			String	150
Respondent			String	150
Exparte			String	150
Petitioner			String	150
Assignee			String	150
Chargee			String	150
Plaintiff			String	150
Applicant			String	150
Registered Owner			String	150
Nature of Business			Text	
Gazette			String	50
Gazette Order			String	50
Strike Off Gazette			String	50
Gazette Notice			String	50
Gazette Date	Gazette date	Format: dd-MM-yyyy	Date	
Gazette Petition			String	50
Gazette Discharge			String	50
Strike Off Date	Strike off date	Format: dd-MM-yyyy	Date	
Date of Letter Section 308		Format: dd-MM-yyyy	Date	
Date of Notice Section 308		Format: dd-MM-yyyy	Date	
Independent Advisor			String	150
Reference			String	50
Auction Date		Format: dd-MM-yyyy	Date	
Hearing Date		Format: dd-MM-yyyy	Date	
Action Date		Format: dd-MM-yyyy	Date	
Notice Date		Format: dd-MM-yyyy	Date	
Petition Date		Format: dd-MM-yyyy	Date	
Order Date		Format: dd-MM-yyyy	Date	
Originating Summon Date		Format: dd-MM-yyyy	Date	
Source			String	50
Discharge Date		Format: dd-MM-yyyy	Date	
Date Incorporated		Format: dd-MM-yyyy	Date	
Annual Return Date		Format: dd-MM-yyyy	Date	
Date of assignment	Date of loan assignment	Format: dd-MM-yyyy	Date	
Sale Order Date		Format: dd-MM-yyyy	Date	

Field	Explanation	Notes	Data Type	Length
Outstanding amount			Decimal	15,2
Shareholding			Integer	15
Amount			Decimal	15,2
Reserved Price			Decimal	15,2
Capital			Integer	15
Paidup			Integer	15
Remark			String	255
Remark 1			String	255
Remark 2			String	255
Remark 3			String	255
Remark 4			String	255
Administrator 1			String	150
Administrator 2			String	150
Administrator 3			String	150
Administrator 4			String	150
Lawyer Name			String	150
Lawyer Address 1			String	150
Lawyer Address 2			String	100
Lawyer Address 3			String	100
Lawyer Address 4			String	100
Lawyer Telephone			String	50
CED Contact Name			String	150
CED Contact Address 1			String	150
CED Contact Address 2			String	100
CED Contact Address 3			String	100
CED Contact Address 4			String	100
CED Contact Telephone			String	50
CED Contact Reference			String	50
Liquidator Name			String	150
Liquidator Address 1			String	150
Liquidator Address 2			String	100
Liquidator Address 3			String	100
Liquidator Address 4			String	100
Liquidator Telephone			String	50
Auctioneer Name			String	150
Auctioneer Address 1			String	150
Auctioneer Address 2			String	100

Field	Explanation	Notes	Data Type	Length
Auctioneer Address 3			String	100
Auctioneer Address 4			String	100
Auctioneer Telephone			String	50
Additional Contact Name			String	150
Additional Contact Address 1			String	150
Additional Contact Address 2			String	100
Additional Contact Address 3			String	100
Additional Contact Address 4			String	100
Additional Contact Telephone			String	50
Settlement Code			String	100
Settlement Date		Format: dd-MM-yyyy	Date	
Settlement Source			String	50
Settlement Source Date		Format: dd-MM-yyyy	Date	
Latest Status Code			String	100
Latest Status Date		Format: dd-MM-yyyy	Date	
Latest Status Source			String	50
Latest Status Source Date		Format: dd-MM-yyyy	Date	
Date Status Expire		Format: dd-MM-yyyy	Date	
Date Status Updated		Format: dd-MM-yyyy	Date	
Other Defendant Name			String	150
Other Defendant Old IC			String	12
Other Defendant New IC			String	20
Director Name			String	150
Director Old IC/Local Number			String	12
Director New IC			String	20
Director Address			String	200
Director SSM Source Date		Format: dd-MM-yyyy	Date	
Director Shares			Integer	15
Director Date Appointed		Format: dd-MM-yyyy	Date	
Director Date Resigned		Format: dd-MM-yyyy	Date	
Director Remark	Director CTOS Remark		String	255

Field	Explanation	Notes	Data Type	Length
Comments from Subject			String	3000
Comments from CRA			String	3000
Comment Date		Format: dd-MM-yyyy	Date	
Withheld Message			String	200

8.6 Section D2

Section D2 is a subsection to Section D which provides information on legal cases such as summons, writs, bankruptcy proceedings, foreclosure and others where the subject is the plaintiff of the action.

Not applicable for method "requestLite"

Content of this section is equivalent to Section D.

8.7 Section D4

Section D4 is a subsection to Section D which provides information specifically on non-personal capacity legal cases.

Not applicable for method "requestLite"

Content of this section is equivalent to Section D.

8.8 Section CCRIS

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  </application>
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    total_outstanding="[Total Outstanding]"
    total_fec="[Total FEC]">
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      [Total Borrowed Outstanding]
    </borrower>
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Guarantor Limit]">
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  <special_attention status="[ Summary Special Attention Status]">
  <special_name status="[ Summary Special Name Status]">
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    <facility code="[Facility Code]">[Facility Description]</facility>
  </application>
  <facilities>
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</facilities>
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</sub_accounts>
</special_attention_acc>
</special_attention_accs>

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Field	Explanation	Notes	Data Type	Length
Entity Key			Integer	11
Entity Warning		Possible values: • Rejected • Pending verification • Wound-Up	String	20
FI Code			String	5
Count Approved	Summary section		Integer	3
Total Approved Amount	Summary section		Decimal	15,2
Count Pending	Summary section		Integer	3
Total Pending Amount	Summary section		Decimal	15,2
Total Limit	Summary section		Decimal	15,2
Total Outstanding	Summary section		Decimal	15,2
Total FEC	Summary section		Decimal	15,2
Total Borrowed	Summary section		Decimal	15,2
Total Borrowed FEC	Summary section		Decimal	15,2
Total Borrowed Outstanding	Summary section		Decimal	15,2
Total Guarantor FEC	Summary section		Decimal	15,2
Total Guarantor Limit	Summary section		Decimal	15,2
Summary Legal Status	Summary section	Possible values: 0 – No record 1 – Has record	Integer	1
Summary Special Attention Status	Summary section	Possible values: 0 – No record 1 – Has record	Integer	1

Field	Explanation	Notes	Data Type	Length
Summary Special Name Status	Summary section	Possible values: 0 – No record 1 – Has record	Integer	1
Application Date		Format: dd-MM-yyyy	Date	
Facility Code	As per BNM return		String	15
Facility Description	As per BNM return		String	60
Total secured facilities			Integer	5
Total unsecured facilities			Integer	5
Total limit %			Integer	5
Average in arrears			Decimal	5,2
Outstanding amount			Decimal	15,2
Credit Card Average Utilisation Last 6 Months	Average utilisation of credit card for the past 6 months	The value represents in percentage	Decimal	5,2
Other Revolving Credits Average Utilisation Last Months	Average utilisation of other revolving credits for the past 6 months	The value represents in percentage	Decimal	5,2
Charge Card Min 12 Months Usage	Lowest amount usage of charge card for the past 12 months		Integer	10
Charge Card Max 12 Months Usage	Highest amount usage of charge card for the past 12 months		Integer	10
Ptptn No Of Account	Number of having PTPTN account by the subject		Integer	4
Local Lender No Of Account	Number of having local lender account by the subject		Integer	4
Foreign Lender No Of Account	Number of having foreign lender account by the subject		Integer	4
Reporting FI Code	Reporting FI for the account.	Possible values: PT = PTPTN Default is Empty value	String	5
Reporting Branch FI Code	Reporting FI branch for the account	Default is empty value	String	5
Application ID			Integer	12
Application date		Format: dd-MM-yyyy	Date	
Application Type		Possible values I – Increase in Limit N – New R – Rollover/Renewal	String	5
Application Status Code		Possible values: P – Pending Decision	String	5

Field	Explanation	Notes	Data Type	Length
		A – Approved R – Rejected X – Cancelled D – Deleted T – Approved by FI, Pending Customer Acceptance		
Application status	Refer to Application Status Code		String	50
Capacity Code		Possible values: Own – Own Sole – Sole proprietor Partner – Partnership Joint – Joint-account Guarantor – Financial guarantee Prof – professional Bodies	String	10
Capacity Desc	Refer to Capacity Code		String	100
Lender Type Code		Possible values: CB – Commercial Bank FC – Finance company MB – Merchant Bank RCB – Related commercial bank RFC – Related finance company RMB – Related merchant bank OTH – Others OWN – Own institution	String	10
Lender Type Desc	Refer to Lender Type Code		String	80
Amount	Approval Limit in RM		Decimal	15,2
Financial Group Resident Status		Possible values: • M • F	String	1
Property Status – Code		Possible values: 100 – Primary completed 101 – Primary under construction 102 – Primary others 200 - Secondary/ sub-sale	String	5
Property Status - Desc	Refer to Property Status – Code		String	28
Property Address Line 1	Property Address Line 1		String	40
Property Address Line 2	Property Address Line 2		String	40
Property Address Line 3	Property Address Line 3		String	40
Property Address Line 4	Property Address Line 4		String	40
Property Address City	Property Address City		String	20
Property Postcode	Property Postcode		Integer	5

Field	Explanation	Notes	Data Type	Length
Property State - Code	Property State Code		String	2
Property State – Desc	Refer to section 16		String	32
Property Country - Code	Property Country Code		String	2
Property Country - Desc	Refer to section 17		String	46
Account Master Data ID	Account Master Data ID		Integer	12
Approval Date		Format: dd-MM-yyyy	Date	
Collateral Code		Possible values: 00 - clean 10- property 21- quoted shares 22- unquoted shared 23- unit trusts 29- other financial assets 30- motor vehicle 40- other vehicle 50- concession & contractual rights 60- plant & machinery 70- financial guarantee 80- negative pledge 90- other assets 99- secured (details unknown)	String	5
Collateral Description	Refer to Collateral Code		String	100
Collateral Value			Decimal	15,2
Latest Status Code		Possible values: 10- summon/writ filed 11- judgement order 12- bankruptcy 13- charging order 14- garnishee order 15- writ of seizure & sale 16- prohibitor order 17- winding-up 18- auction 19- judgement debtor summon 20- receiver/Section 176 21- settled/discharged	String	5
Latest Status Description	Refer to Latest Status Code		String	50
Latest Update Date	Latest legal update date	Format: dd-MM-yyyy	Date	
Facility – Code	As per BNM return		String	15
Facility – Desc	As per BNM return		String	60
Repay - Code		Possible values: ODD – On Demand DLY – Daily WKY – Weekly FNY – Fortnightly	String	15

Field	Explanation	Notes	Data Type	Length
		MTH – Monthly QTR – Quarterly HYR – Semi/half yearly ANN – Annually BUL – Bullet REV – Revolving IDF – Indefinite IRR – Irregular schedule		
Repay - Desc	Refer to Repay. Code		String	100
Position Code	Position status	Possible values: C – Rescheduled credit facility G – Outstanding from call of guarantee O – Outstanding T – Restructured S – Settled P – Partial W – Written Off K – Loan under AKPK's Assisted Repayment Programmes, such as Debt Management Programme (DMP) and Small Debt Resolution Scheme (SDRS). J -Loan under AKPK's Interim Repayment Assistance Programmes, such as Financial Management and Resilience Programme (URUS) and Financial Resilience Support Scheme (FIRST) L -Impaired Loan/Financing Sold to Third Party2	String	5
Position Description	Refer to Position Code		String	50
Position Balance			Decimal	15,2
Position Date		Format: dd-MM-yyyy	Date	
No. of Instalment			Integer	5
Months in Arrears			Integer	5
Instalment Amount			Decimal	15,2
Rescheduled Date	When Position Code is C (Rescheduled credit facility)	Format: dd-MM-yyyy	Date	
Restructured Date	When Position Code is T (Restructured)	Format: dd-MM-yyyy	Date	
Application Data Id			Integer	11

8.9 Section DCHEQS

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</dcheque_owns>

```

Field	Explanation	Notes	Data Type	Length
Bank Number	Bank Number from DCHEQS		String	5
Account Number	Account number		String	10
Date Issued	Issuance Date	Format: dd-MM-yyyy	Date	
Remarks	Remarks from DCHEQS		String	50
Bank Code	Bank's FI Code		String	5
Bank Name	Name of bank		String	100
Cheque Number	Cheque number		String	50
Amount	Amount in cheque		Decimal	15,2

8.10 Section CCRIS SUPPLEMENTARY

```

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    <address6>[Address 6]</address6>
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    <state_code>[State Code]</state_code>
    <country>[Country]</country>
    <post_code>[Post Code]</post_code>
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    <branch_code="[ Reporting Branch FI Code]" />
    <application_code="[ Reporting Application FI Code]" />
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</addresses>
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<application_code="[ Reporting Application FI Code]" />
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</contact>
</contacts>
<occupations>
  <occupation>
    <occ_type>[Occupation Type]</occ_type>
    <occupation>[Occupation]</occupation>
    <employer>[Employer]</employee>
    <from_date>[From Date]</from_date>
    <to_date>[To Date]</to_date>
    <org_code="[Reporting FI Code]" />
    <branch_code="[ Reporting Branch FI Code]" />
    <application_code="[ Reporting Application FI Code]" />
    <create_date>[Occupation Create Date]</create_date>
  </occupation>
</occupations>

```

Field	Explanation	Notes	Data Type	Length
Address				
Address 1	Address Line 1		String	50
Address 2	Address Line 2		String	50
Address 3	Address Line 3		String	50
Address 4	Address Line 4		String	50
Address 5	Address Line 5		String	50
Address 6	Address Line 6		String	50
Long Address	Long Address		String	150
State Code	State Code		String	6
Country	ISO Country Description		String	80
Post Code	Post Code		String	10
Reporting FI Code	Reporting FI for the account.		String	5
Reporting Branch FI Code	Reporting FI branch for the account		String	5
Reporting Application FI Code	Reporting FI Application Code		String	3
Address Create Date	Address Creation Date	Format: dd-MM-yyyy	Date	
Contact				
Contact Type	Contact Type		String	3
ISD Code	ISD Code		Integer	5
Prefix	Prefix		Integer	7

Field	Explanation	Notes	Data Type	Length
Phone Number	Phone Number		String	25
Contact Create Date	Contact Creation Date	Format: dd-MM-yyyy	Date	
Occupation				
Occupation Type	Occupation Type		String	3
Occupation	Occupation		String	80
Employer	Employer		String	50
From Date	From Date	Format: dd-MM-yyyy	Date	
To Date	To Date	Format: dd-MM-yyyy	Date	
Occupation Create Date	Occupation Creation Date	Format: dd-MM-yyyy	Date	

8.11 Section ANGKASA

```

<information>
  <matched_name>[Name]</matched_name>
  <matched_new_ic>[New IC]</matched_new_ic>
  <matched_old_ic>[Old IC]</matched_old_ic>
  <status>[Status]</status>
  <remark>[Remark]</remark>
  <reference_no>[Reference No.]</reference_no>
  <salary_no>[Salary No.]</salary_no>
</information>
<monthly_deduction seq="[Sequence]">
  <itemise>
    <item>
      <org_name>[Org. Name]</org_name>
      <org_no>[Org. No.]</org_no>
      <org_status>[Org. Status]</org_status>
      <org_deduction_amount>[Org. Deduct]</org_deduction_amount>
      <details>
        <detail>
          <type>[Type]</type>
          <begin>[Begin Date]</begin>
          <end>[End Date]</end>
          <deduction_amount>[Deduct Amt.]</deduction_amount>
        </detail>
      </details>
    </item>
  </itemise>
  <total_deduction>[Total Deduct]</total_deduction>
</monthly_deduction>
<pending_deduction seq="[Sequence]">
  <details>
    <detail>
      <org_name>[Org. Name]</org_name>
      <org_no>[Org. No.]</org_no>
      <org_status>[Org. Status]</org_status>
      <start_date>[Begin Date]</start_date>
    </detail>
  </details>
</pending_deduction>

```

```

        <deduction_amount>[Deduct Amt.]</deduction_amount>
    </detail>
</details>
    <total_pending_deduction>[Total Pending
Deduct]</total_pending_deduction>
</pending_deduction>
<payment_history seq="[Sequence]">
    <payment_month>
        <item>[Payment Date]</item>
    </payment_month>
    <payment_detail>
        <item>
            <code>[Code]</code>
            <process_date>
                <item>[Process Date]</item>
            </process_date>
            <amount>
                <item>[Amount]</item>
            </amount>
        </item>
    </payment_detail>
</payment_history>

```

Field	Explanation	Notes	Data Type	Length
Information				
Name	Name from Angkasa		String	255
New IC	New IC from Angkasa		String	50
Old IC	Old IC from Angkasa		String	50
Status	Status from Angkasa	e.g. Active	String	50
Remark	Remark from Angkasa		String	255
Reference No.	Reference Number from Angkasa		String	255
Salary No.	Salary Number from Angkasa	e.g. 08238602	String	50
Monthly Deduction				
Sequence	CTOS sequence number	Running number	Integer	5
Org. Name	Organization name		String	50
Org. No.	Organization number		String	50
Org. Status	Organization Status	e.g. Active	String	25
Org. Deduct	Organization Deduction Amount	e.g. 150.00	Decimal	15,2
Total Deduct	Total deduction	e.g. 150.00	Decimal	15,2
Detail				
Type	Facility Type	e.g. Simpanan Khas (0301)	String	50
Begin Date	Date Begin	Format: MM/yyyy e.g. 01/2018	Date	
End Date	Date End	Format: MM/yyyy e.g. 12/2019	Date	
Deduct. Amt.	Deduction amount	e.g. 150.00	Decimal	15,2

Field	Explanation	Notes	Data Type	Length
Pending Deduction				
Sequence	CTOS sequence number	Running number	Integer	5
Total Pending Deduct	Total Pending deduction	e.g. 150.00	Decimal	15,2
Detail				
Org. Name	Organization name		String	50
Org. No.	Organization number		String	50
Org. Status	Organization Status	e.g. Active	String	25
Begin Date	Date Begin	Format: MM/yyyy e.g. 01/2018	Date	
Deduct Amt.	Total deduction	e.g. 150.00	Decimal	15,2
Payment History				
Sequence	CTOS sequence number	Running number	Integer	5
Payment Month				
Payment Date	Payment Date (12 payment date)	Format: MMM yy e.g. Sep 17	Date	
Payment Details				
Code	Organization Code	e.g. BL504	String	50
Process Date	Date processed <i>Note: "-" is returned if no data available.</i>	Format: Dd MMM yy e.g. 24 Feb 17	Date	
Amount	Amount paid <i>Note: "-" is returned if no data available.</i>	e.g. 150.00	Decimal	15,2

8.12 Section ETR PLUS

```

<etrPlusSummary>
  <creditAccount>[Credit Account]</creditAccount>
  <active>[Active]</active>
  <outStanding>[Outstanding]</outStanding>
  <closed>[Closed]</closed>
  <outStandingOverDue>[Outstanding Overdue]</outStandingOverDue>
  <prevOutStandingOverDue>[Previous Outstanding
Overdue]</prevOutStandingOverDue>
  <avgDueDays>[Average Due Days]</avgDueDays>
  <ontimePayPercentage>[On Time Pay Percentage]</ontimePayPercentage>
</etrPlusSummary>
<etrPlusAccountProfileList>
  <etrPlusAccountProfile>
    <industry>[Industry]</industry>
    <account>[Account]</account>
    <paymentTerm>[Payment Term]</paymentTerm>
    <asAtDate>[As at Date]</asAtDate>
    <amountDue>[Amount Due]</amountDue>
    <withinTerm>[Within Term]</withinTerm>
    <age1to30>[Age 1 to 30]</age1to30>
  </etrPlusAccountProfile>
</etrPlusAccountProfileList>

```



```
<age31to60>[Age 31 to 60]</age31to60>
<age61to90>[Age 61 to 90]</age61to90>
<age91to120>[Age 91 to 120]</age91to120>
<ageOver120>[Age Over 120]</ageOver120>
</etrPlusAccountProfile>
</etrPlusAccountProfileList>
<etrPlusPaymentHistoryList>
  <etrPlusPaymentHistory>
    <industry>[Industry]</industry>
    <account>[Account]</account>
    <paymentTerm>[Payment Term]</paymentTerm>
    <capacity>[Capacity]</capacity>
    <monthDetailsList>
      <monthDetails>
        <month>[Month]</month>
        <year>[Year]</year>
        <payment>[Payment]</payment>
      </monthDetails>
    </monthDetailsList>
  </etrPlusPaymentHistory>
</etrPlusPaymentHistoryList>
<etrPlusNBLHistory>
  <totalLimit>[Total Limit]</totalLimit>
  <overallTotalOutstandingBalance>[Overall Total Outstanding
Balance]</overallTotalOutstandingBalance>
  <etrPlusNBLCreditHistoryList>
    <etrPlusNBLCreditHistory>
      <number>[Number]</number>
      <approvalDate>[Approval Date]</approvalDate>
      <capacity>[Capacity]</capacity>
      <lenderType>[Lender Type]</lenderType>
      <facility>[Facility]</facility>
      <limit>[Limit]</limit>
      <installmentAmount>[Installment
Amount]</installmentAmount>
      <tenure>[Tenure]</tenure>
      <dateBalanceUpdated>[Date Balance
Updated]</dateBalanceUpdated>
      <totalOutstandingBalance>[Total Outstanding
Balance]</totalOutstandingBalance>
      <principalRepaymentTerm>[Principal Repayment
Term]</principalRepaymentTerm>
      <collateralType>[Collateral Type]</collateralType>
      <status>[Status]</status>
      <dateStatusUpdated>[Date Status Updated]</
dateStatusUpdated >
      <monthDetailsList>
        <monthDetails>
          <month>[Month]</month>
          <year>[Year]</year>
          <payment>[Payment]</payment>
        </monthDetails>
      </monthDetailsList>
    </etrPlusNBLCreditHistory>
  </etrPlusNBLCreditHistoryList>
</etrPlusNBLHistory>
```

Field	Explanation	Notes	Data Type	Length
EtrPlusSummary				

Field	Explanation	Notes	Data Type	Length
Credit Account	Total number of accounts		Integer	5
Active	Total number of active accounts		Integer	5
Outstanding	Total number of accounts which have outstanding amount		Integer	5
Closed	Total number of closed accounts		Integer	5
OutStanding OverDue	Total outstanding amount for all accounts		Double	12,2
Previous Outstanding OverDue	Total outstanding amount of all accounts		Double	12,2
Average Due Days	Average of credit terms of all accounts		Double	3,1
On Time Pay Percentage	Percentage of payment made on time without fail		Double	3,1
EtrPlusAccountProfileList				
EtrPlusAccountProfile				
Industry	Industry		String	255
Account	Account no		String	30
Payment Term	Credit term in No of days		Integer	3
As at Date	Statement date of the account	Format: yyyy-MM-dd	Date	
Amount Due	Total amount due mentioned as a range		Double	12,2
Within Term	Percentage of Current (RM) of an account (age30)		Double	12,2
Age 1 to 30	Percentage of aging between 1-30 days after credit term Over payment over total payment		Double	12,2
Age 31 to 60	Percentage of aging between 31-60 days after credit term Over payment over total payment		Double	12,2
Age 61 to 90	Percentage of aging between 61-90 days after credit term Over payment over total payment		Double	12,2
Age 91 to 120	Percentage of aging between 91-120 days after credit term Over payment over total payment		Double	12,2
Age Over 120	Percentage of aging above 120 days after credit term Over payment over total payment		Double	12,2
EtrPlusPaymentHistoryList				
EtrPlusPaymentHistory				
Industry	Industry		String	255
Account	Account no		String	30

Field	Explanation	Notes	Data Type	Length
Payment Term	Credit term in No of days		Integer	3
Capacity	Capacity	Possible values: Personal: For direct subject Non-Personal: For sponsor	String	20
MonthDetailsList				
MonthDetails				
Month	Month	Eg., Jan - Dec	String	3
Year	Year		Integer	4
Payment		Eg., 0, 1, 2, 3, 4, 5, 6, 6+ Note: It may show below code: s: Fully Settled w: Written Off Bad Debt	String	2
EtrPlusNBLHistory				
Total Limit	Total limit of all the accounts		Double	12,2
Overall Total Outstanding Balance	Added sum of total outstanding amount of all the accounts		Double	12,2
EtrPlusNBLCreditHistoryList				
EtrPlusNBLCreditHistory				
Number	Sequence no		Integer	5
Approval Date	Approval Date	Format: YYYY-MM-dd	Date	
Capacity	Capacity	Possible values: P: Personal: For direct subject NP: Non-Personal: For sponsor	String	20
Lender Type	Sub sector of the Company which submitted the data	Possible values: BNPL: Buy Now Pay Later RE: Real Estate	String	255
Facility	Facility of the latest submitted aging data	Possible values: TF: Trade Facilities GS: Goods & Services Rendered TL: Term Loan ALR: Agriculture Land Rental CR: Commercial Rental DLR: Development Land Rental IR: Industrial Rental	String	25

Field	Explanation	Notes	Data Type	Length
		ISM: Instalment RR: Residential Rental		
Limit	Latest Account limit		Double	12,2
Installment Amount	Latest Instalment amount of the account		Double	12,2
Tenure	Latest tenure in number of months		Integer	5
Date Balance Updated	Latest Statement date of the account	Format: YYYY-MM-dd	Date	
Total Outstanding Balance	Latest total outstanding amount of the account		Double	12,2
Principal Repayment Term	Latest principal payment term in Abbreviation of the Account		String	25
Collateral Type	Latest collateral type of the account in Abbreviation	Possible values: CONC: Concession And Other Contractual Rights DPS: Deposit FG: Financial Guarantee MTV: Motor Vehicles (Registered With JPJ) NP: Negative Pledge OAS: Other Assets OFA: Other Financial Assets OTV: Other Vehicles/Carriers PAM: Plat And Machinery PROP: Properties QSHA: Quoted Shares SEC: Secured- Details Unknown SL: Supportive Letters UNQSHA: Unquoted Shares UNT: Unit Trusts	String	25
Status	Latest legal status of the account	EG., LEGAL, NPL	String	25
Date Status Updated	Latest date of status updated	Format: YYYY-MM-dd	Date	
MonthDetailsList				
MonthDetails				
Month	Month	Eg., Jan - Dec	String	3
Year	Year		Integer	4

Field	Explanation	Notes	Data Type	Length
Payment		Eg., 0, 1, 2, 3, 4, 5, 6, 6+ Note: It may show below code: - s - Fully Settled - w - Written Off Bad Debt	String	2

8.13 Section E

```

<enquiry seq="[Sequence]" account_no="[Account No]" tref_id="[Tref Id]">
  <subject>
    <ref_com_name>[Reference Company Name]</ref_com_name>
    <ref_com_bus>[Reference Company business]</ref_com_bus>
    <party_type>[Subject Party Type]</party_type>
    <ic_lcno>[Subject Old IC]</ic_lcno>
    <nic_brno>[Subject New iC]</nic_brno>
    <name>[Subject Name]</name>
    <address>[Address]</address>
    <tref_date>[Tref Date]</tref_date>
  </subject>
  <relationship>
    <rel_type code="[Relationship Type Code]">[Relationship Type
Desc]</rel_type>
    <rel_status code="[Relationship Status Code]">[Relationship
Status Desc]</rel_status>
    <account_no>[Account No]</account_no>
    <rel_syear>[Relationship Start Year]</rel_syear>
    <rel_smonth>[Relationship Start Month]</rel_smonth>
    <rel_sday>[Relationship Start Day]</rel_sday>
    <remark>[Relationship Remark]</remark>
  </relationship>
  <sponsor>
    <ref_com_name>[Reference Company Name]</ref_com_name>
    <ref_com_bus>[Reference Company business]</ref_com_bus>
    <ic_lcno>[Sponsor Old IC]</ic_lcno>
    <nic_brno>[Sponsor New iC]</nic_brno>
    <sp_party_type>[Sponsor Party Type]</sp_party_type>
    <name>[Sponsor Name]</name>
    <date>[Sponsor Date]</date>
    <sp_type code="[Sponsor Type Code]">[Sponsor Type Desc]</sp_type>
    <guarantor>[Is Guarantor]</guarantor>
    <sp_source code="[Sponsor Source Code]">[Sponsor Source
Desc]</sp_source>
    <sp_alias>[Sponsor Alias]</sp_alias>
  </sponsor>
  <account_status>
    <account_no>[Account No]</account_no>
    <statement_date>[Statement Date]</statement_date>
    <account_rating code="[Rating of Account Code]">[Rating of
Account Desc]</account_rating>
    <account_term>[Account Term]</account_term>
    <account_limit>[Account Limit]</account_limit>
    <account_status code="[Account Status Code]">[Account Status
Desc]</account_status>
    <account_ref_no>[Account Reference No]</account_ref_no>
    <debtor_name>[Debtor Name]</debtor_name>

```

	debtor_ic_lcnno	Debtor	OLD	IC/Company	Local
No]	</debtor_ic_lcnno>				
	<debtor_nic_brno>[Debtor New IC/Business No]</debtor_nic_brno>				
	<address>[Debtor Address]</address>				
	<debt_type code="[Debt Type Code]">[Debt Type Desc]</debt_type>				
	<last_paid_amount>[Last Paid Amount]</last_paid_amount>				
	<age>				
	<age_30>[30 Days aging]</age_30>				
	<age_60>[60 Days aging]</age_60>				
	<age_90>[90 Days aging]</age_90>				
	<age_120>[120 Days aging]</age_120>				
	<age_150>[150 Days aging]</age_150>				
	<age_180>[180 Days aging]</age_180>				
	<age_over_180>[210 Days aging]</age_over_180>				
	</age>				
	</account_status>				
	<return_cheques status="[Status]">				
	<record seq="[Sequence]">				
	<cheque_no>[Cheque No]</cheque_no>				
	<cheque_acc_no>[Bank Account No]</cheque_acc_no>				
	<cheque_bank code="[Bank Code]">[Bank Desc]</cheque_bank>				
	<cheque_reason code="[Reason Code]">[Reason				
	Desc]</cheque_reason>				
	<cheque_issuer code="[Check Issuer Code]">[Check Issuer				
	Desc]</cheque_issuer>				
	<cheque_amount>[Cheque Amount]</cheque_amount>				
	<cheque_date>[Cheque Date]</cheque_date>				
	</record>				
	</return_cheques>				
	<legal_action status="[Status]">				
	<reminder_letter>				
	<title>[Legal Action Type]</title>				
	<date>[Legal Action Date]</date>				
	<reference>[Comment]</reference>				
	</reminder_letter>				
	<demand_letter_by_company>				
	<title>[Legal Action Type]</title>				
	<date>[Legal Action Date]</date>				
	<reference>[Comment]</reference>				
	</demand_letter_by_company>				
	<demand_letter_by_lawyer>				
	<title>[Legal Action Type]</title>				
	<date>[Legal Action Date]</date>				
	<lawyer>[Remark]</lawyer>				
	<reference>[Comment]</reference>				
	</demand_letter_by_lawyer>				
	</legal_action>				
	<cra_comment>				
	<cra_status_id>[CRA Status ID]</cra_status_id>				
	<cra_status>[CRA Status]</cra_status>				
	<cra_status_desc>[CRA Status Desc]</cra_status_desc>				
	<cra_comment>[CRA Comment]</cra_comment>				
	</cra_comment>				
	<subject_comments>				
	<subject_comment>[Subject Comment]</subject_comment>				
	</subject_comments>				
	<referee_contact>				
	<contact_ref>[Contact Reference]</contact_ref>				
	<contact_name>[Contact Name]</contact_name>				
	<contact_add>[Contact Address]</contact_add>				
	<contact_telno>[Contact Tel No]</contact_telno>				

```

    <contact_nature_of_business>[Contact Nature of
Business]</contact_nature_of_business>
    <contact_faxno>[Contact Fax No]</contact_faxno>
    <contact_email>[Contact Email]</contact_email>
    <contact_type code=" [Contact Type Code]">[Contact Type
Desc]</contact_type>
    </referee_contact>
</enquiry>

```

Field	Explanation	Notes	Data Type	Length
Enquiry				
Sequence	Sequence no		String	3
Account No	Account number of that relationship		String	30
Tref Id	Tref ID of the Subject		String	11
Subject				
Reference Company Name	Subject's ref company name		String	255
Reference Company business	Business of subject's ref company		String	255
Subject Party Type	Subject's party type		String	1
Subject Old IC/Company Local No.	Old IC/Company Local No. of the subject		String	20
Subject New IC/Business No.	New IC/Business No. of the subject		String	20
Subject Name	Name of the subject		String	200
Address	Address of the subject		String	255
Tref Date	Trade reference Date	Format: yyyy-MM-dd	date	
Relationship				
Relationship Type Code	Code of relationship type		String	6
Relationship Type Desc	Description of relationship type		String	50
Relationship Status Code	Code of relationship status		String	6
Relationship Status Desc	Description of relationship status		String	30
Account No	Account number		String	30
Relationship Start Year	Relationship start year		String	6

Field	Explanation	Notes	Data Type	Length
Relationship Start Month	Relationship start month		String	6
Relationship Start Day	Relationship start day		String	6
Relationship Remark	Relationship remark		String	255
Sponsor				
Reference Company Name	Sponsor's ref company name		String	255
Reference Company business	Business of sponsor's ref company		String	255
Sponsor Old IC/Company Local No.	Old IC/Company Local No. of the sponsor		String	20
Sponsor New IC/Business No.	New IC/Business No. of the sponsor		String	20
Sponsor Party Type	Sponsor's party type		String	1
Sponsor Name	Name of the sponsor		String	200
Sponsor Date	Date of sponsor	Format: yyyy-MM-dd	Date	
Sponsor Type Code	Code of sponsor type		String	10
Sponsor Type Desc	Description of sponsor type		String	50
Is Guarantor	Is Guarantor		Integer	1
Sponsor Source Code	Code of sponsor source		String	10
Sponsor Source Desc	Description of sponsor source		String	50
Sponsor Alias	Alias of sponsor		String	50
Account Status				
Account No	Account number		String	30
Statement Date	Statement date	Format: yyyy-MM-dd	Date	
Rating of Account Code	Code of account rating		String	6
Rating of Account Desc	Description of account rating		String	50
Account Term	Term period for the account		String	6
Account Limit	Credit limit of account		String	11

Field	Explanation	Notes	Data Type	Length
Account Status Code	Code of account status		String	6
Account Status Desc	Description of account status		String	30
Account Reference No	Account Reference Number		String	30
Debtor Name	Name of debtor		String	200
Debtor Old IC/Company Local No	Old IC/Company Local No of debtor		String	20
Debtor New IC/Business No	New IC/Business No of debtor		String	20
Debtor Address	Address of debtor		String	255
Debt Type Code	Code of debtor type		String	6
Debt Type Desc	Description of debtor type		String	50
Last Paid Amount	Account aging last paid amount		Decimal	12,2
30 Days Aging	Aging amount for 1 -30 days		Decimal	12,2
60 Days Aging	Aging amount for 31 – 60 days		Decimal	12,2
90 Days Aging	Aging amount for 61 – 90 days		Decimal	12,2
120 Days Aging	Aging amount for 91 – 120 days		Decimal	12,2
150 Days Aging	Aging amount for 121 – 150 days		Decimal	12,2
180 Days Aging	Aging amount for 151 – 180 days		Decimal	12,2
210 Days Aging	Aging amount for 181 – 210 days		Decimal	12,2
Return Cheque				
Status	Status of the return cheque		String	50
Sequence	Sequence No		String	3
Cheque No	Cheque no		String	6
Cheque Account No	Cheque account no		String	20
Bank Code	Code of bank name		String	10
Bank Desc	Description of bank name		String	50
Reason Code	Code of cheque issued reason		String	6

Field	Explanation	Notes	Data Type	Length
Reason Desc	Description of cheque issued reason		String	50
Cheque Issuer Code	Code of cheque issued		String	6
Cheque Issuer Desc	Description of cheque issued		String	20
Cheque Amount	Cheque amount		Decimal	12,2
Cheque Date	Cheque date	Format: yyyy-MM-dd	Date	
Legal Action				
Status	Status of legal action		String	50
Legal Action Type	Gives the title of Legal action		String	50
Legal Action Date	Gives the date on which Legal action was taken	Format: yyyy-MM-dd	Date	
Comment	Gives the comments if any		String	100
Remark	Lawyer's remark		String	50
CRA Comment				
CRA Status ID	CRA status ID		String	11
CRA Status	CRA status		String	200
CRA Status Desc	Description of CRA status		String	255
CRA Comment	CRA comment		String	255
Subject Comment				
Subject Comment	Subject comment		Text	
Contact				
Contact Reference	Contact reference		String	30
Contact Name	Contact person's name		String	50
Contact Address	Contact address		String	255
Contact Tel No	Contact tel no		String	20
Contact Nature of Business	Nature of contact person's business		String	255
Contact Fax No	Contact fax no		String	20
Contact Email	Contact email		String	100
Contact Type Code	Code of contact type		String	6
Contact Type Desc	Description of contact type		String	200

9 XML to HTML Formatting Using Stylesheet

A XSLT stylesheet will be provided for proper formatting of the XML output file into HTML that can be displayed by popular web browsers like Internet Explorer, Mozilla Firefox, Google Chrome, Safari and others. Thus, users need not worry about the details of the XML output file format.

The stylesheet is a combined stylesheet which support both kinds of request methods:

- request – returns CTOS Report
- requestLite – returns Lite Report

For each method, the template base is different as below:

- request - ctos_report.xsl
- requestLite - lite_report.xsl

Please take note of the respective base stylesheet corresponding to the request method in order to correctly display the report in HTML.

In the event the SI requires parsing the entire XML output to extract the contents of data items, this should be done with care to prevent extracting and processing incorrect data. In the event assistance is required, the SI may contact the personnel on the contact list.

10 Integration Connectivity

10.1 UAT Environment

The UAT environment will require a set of authentication credentials where CTOS will provide the authentication credentials based on a request made by the integrator. The authentication credentials provided will contain the following:

- Company Code
- Account
- User ID
- Password
- Expiry Date

Along with the authentication credentials, a list of test data will be provided which will hit Section A, Section B, Section C and Section D combination. The integrator may also request for certain combination test data if not found in the list.

10.2 Production Environment

Once UAT has been completed and ready to-go-live, the integrator will need to request from CTOS a new set of authentication credentials. The integrator will then need to change the authentication credential previously applied during testing and UAT environment:

- Company Code
- Account
- User ID

- Password

11 Summary Section Status Codes

11.1 Successful CTOS Report

The STAT code of the Summary Section of CTOS XML enquiry report consists of a 4-digit string. Each digit represents the status of one particular section of the CTOS report. The left most digit represents Section A, the next digit represents Section B etc.

The meaning of the value of each digit is different depending on which section it is presenting, as shown in the table below.

Status Code	Section			
	A	B	C	D and D2
0	Nothing is found in the corresponding section (-).	Nothing is found in the corresponding section (-).	Nothing is found in the corresponding section (-).	Nothing is found in the corresponding section (-).
1	Subject's name & ID number is verified (a).	Subject is in your monitoring list only with enquiry history (b).	Subject's directorships & business interests – present & past records (c).	
2	There is a search report from CCM (ä).			
5	Inconsistent name and/or ID number (A)	Subject is in your Internal List only or both Internal List & Monitoring List with enquiry history (B).		At least one record in this section bears a match on the ID number provided. Section D and D2. Requires your careful study and probe (D).
6	There is a search report from CCM based on ID number (Ä)			At least one record in this section bears a match on the ID number provided. Section D2 only (D).
8				At least one record in this section bears a match on the ID number provided and at least one record is withheld due

Status Code	Section			
	A	B	C	D and D2
				to verification and update required from the subject (D). Section D and D2.

Section A (Identity Verification). First position of Status field

"-" indicates no data is found in this section.

The XML equivalent is "0"

"a" indicates Subject's name, ID or company number is verified correct.

The XML equivalent is "1"

"ä" indicates that there is a search report from CCM. This character only applies to company searches only, ie the **Type** is "C".

The XML equivalent is "2"

"A" indicates Inconsistent name and/or ID number. Further verification by the users is required.

The XML equivalent is "5"

"Ä" indicates Inconsistent name and/or ID number, and there is a search report from CCM.

Further verification by the users is required

The XML equivalent is "6"

Section B (Internal List & Monitoring List with Enquiry History). Second position of Status field

"-" indicates no data is found in this section.

The XML equivalent is "0"

"b" indicates Subject is in the user's Monitoring List with Enquiry History.

The XML equivalent is "1"

"B" indicates Subject is in the user's Internal List with Enquiry History.

The XML equivalent is "5"

Section C (Directorship and Business Interests). Third position of Status field

"-" indicates no data is found in this section.

The XML equivalent is "0"

"c" indicates Subject has Directorships and/or Business Interests.

The XML equivalent is "1"

Section D (Legal Proceedings) & Section D2 (Subject as a Plaintiff). Forth position of Status field

"-" indicates no data is found in this section.

The XML equivalent is "0"

“D” indicates Subject has some Legal proceedings records (Section D) and at least one or no records where the Subject is a plaintiff (Section D2). At least one record bears a match on the ID number. Further verification by the users is required.

The XML equivalent is “5”

“D̄” indicates Subject has only records where the Subject is a plaintiff (Section D2) without any legal proceeding records (Section D).

The XML equivalent is “6”

“D̄” indicates Subject has some Legal proceedings records. At least one record bears a match on the ID number and at least one record is withheld as verification/update of the status of the record is required by the Subject.

The XML equivalent is “8”

12 List of purposes

Based on the Credit Reporting Agency Act (CRAA), every enquiry must follow with a purpose. CTOS has categorized the purposes as follow:

Code	Purpose
1	Credit evaluation on subject with consent
2	Credit evaluation on director/shareholder with consent
3	Account opening on subject with consent
4	Account Opening on director/shareholder with consent
5	Credit Review on existing customer
6	Debt Recovery on Non-performing account with consent
7	Due diligence on AMLA compliance with consent
8	Employment on subject with consent
9	Vendor evaluation on subject with consent

General purpose code for B2B Integration which will cover all purposes above:

Code	Purpose
200	Credit evaluation/Account opening on subject/ directors/shareholder with consent / Due diligence on AMLA compliance with consent

13 Due Diligent Index

The Due Diligent Index is a scoring system based on four (4) digit numbers known as ‘index’ where the index for each enquiry name is calculated on the fly based on the information from Section D. The Due Diligent Index will return the code of the worst case in Section D.

The higher the index indicates the severity level of the needs to perform due diligence for evaluation purpose.

The Due Diligent Index is broken down into three (3) sections from the 4 digit numbers.

Digit	Description	Notes
First two digit	The first two digit represents the worst case in Section D – based on record type.	Possible values start from 00 to 99 where the higher the number, the severity of the case increases.
Third Digit	The third digit represents how long the worst case in Section D was in the database	Possible values start from 0 to 9 where the higher the number, the more recent the record.
Forth Digit	The forth digit represents how many records are there in Section D including the worst case used to formulate the first three digits.	Possible values start from 0 to 9 where the higher the number, the more cases in Section D.

NOTE: There will be scenario where a Due Diligent Index cannot be calculated in which the index returned will be “---”. This scenario is when the worst case record in Section D is withheld.

13.1 Due Diligent Index Table

13.1.1 Individual

Digit	Description	Record Type
99	ID Series	Subject Deceased
98	ID Series	Mismatch of name versus legal ID
97	ID Series	Mismatch of input ID versus legal ID
95	AO Series	Bankruptcy Receiving or Adjudication Order received
93	CP Series	Bankruptcy Creditors Petition issued
91	BN Series	Bankruptcy Notice issued
89	PS Series	Proclamation of Sale under deed of assignment
88	PS Series	Proclamation of Sale under the National Land Code Foreclosure proceedings – Originating Summons issues
87	CS Series	Civil Suit Summon
85	Tribunal Series	Tribunal
83	Internal Series	Special Listing – AMLA
82	Internal Series	Internal List
49	Settlement Series	Bankruptcy Discharge/Annulment
48	Settlement Series	Bankruptcy Creditors Petition struck off Action withdrawn Settled
47	Settlement Series	Bankruptcy Notice struck off Action withdrawn Settled
45	Settlement Series	Proclamation of Sale struck off Action withdrawn Settled
43	Settlement Series	Civil Suit struck off

Digit	Description	Record Type
		Action withdrawn Settled
37	Settlement Series	Tribunal struck off Action withdrawn Settled
25	Director Series	Director with company winding up order issued
24	Director Series	Director with company winding up petition issued

13.1.2 Company

Digit	Description	Record Type
98	ID Series	Significant mismatch of company name versus registration number
97	ID Series	Company struck off Business deregistered
96	WUO Series	Voluntary Winding -Up
95	WUO Series	Wound up order issued
93	WUP Series	Petition for winding up issued
89	PS Series	Proclamation of sale under the deed of assignment
88	PS Series	Proclamation of sale under the National Land Code Foreclosure proceedings – Originating Summons issued
87	CS Series	Civil Suit Summon
86	CS Series	Origintating Summons
85	Tribunal Series	Tribunal
83	Internal Series	Internal List Special Listing - AMLA
49	Settlement Series	Wound up order struck off Stayed
48	Settlement Series	Petition stayed struck off Action Withdrawn Settled
47	Settlement Series	Proclamation of sale struck off Action withdrawn Settled
45	Settlement Series	Civil Suit struck off Action withdrawn Settled
43	Settlement Series	Civil Suit under repayment scheme
41	Settlement Series	Civil Suit – Account regularized
37	Settlement Series	Company under special administration
35	Settlement Series	Company Restructure/ equity reduction

14 Internal standardization formats

14.1 Standardizations of Subject Name

CTOS system performs name transformations to ensure correct matching of records in our database.

14.1.1 Company name transformation

Input name	Transformed Name
SENDIRIAN BERHAD	SDN BHD
SENDIRIAN	SDN
BERHAD	BHD
PRIVATE LIMITED	PTE LTD
LIMITED	LTD
SDN. BHD.	SDN BHD
SDN.	SDN
BHD.	BHD
PTE. LTD.	PTE LTD
PTE.	PTE
LTD.	LTD

14.1.2 Individual name transformation

Input name	Transformed Name
S/O	A/L
D/O	A/P
BINTE	BTE
BINTI	BTE
BT	BTE
HJ(appears in the middle of a name)	HAJI

The following titles if appear at the beginning of a name, will be removed before matching is performed:
TAN SRI, YB, DATUK, DATOK, DATO, DATO', DATIN, HAJI, HJ, HAJAH, HAJJAH, DR, MR, MRS, MS, MISS

15 Record Type for Section D

Below are the list of record type for Section D

Code	Value
C1	COMPANY WINDING UP
C2	COMPANY SUMMONS / CIVIL SUIT
C4	VOLUNTARY WINDING UP
C5	COMPANY RESTRUCTURE
C6	PROCLAMATION OF SALE UNDER DEED OF ASSIGNMENT
C7	FORECLOSURE PROCEEDINGS ORIGINATING SUMMONS ISSUED / PROCLAMATION OF SALE UNDER THE NATIONAL LAND CODE
C8	PROCLAMATION OF SALE UNDER THE NATIONAL LAND CODE
C9	FORECLOSURE PROCEEDINGS STATUTORY NOTICE ISSUED
CA	PROCLAMATION OF SALE
CC	ORIGINATING SUMMONS
CD	POWER OF REGISTRAR TO STRIKE OFF DEFUNCT COMPANY FROM REGISTER

CF	COMPANY UNDER SPECIAL ADMINISTRATION (PAST & CURRENT)
I1	BANKRUPTCY
I2	SUMMONS / CIVIL SUIT
I4	DIRECTOR OF WOUND UP
I5	PROCLAMATION OF SALE UNDER DEED OF ASSIGNMENT
I6	FORECLOSURE PROCEEDINGS ORIGINATING SUMMONS ISSUED / PROCLAMATION OF SALE UNDER THE NATIONAL LAND CODE
I7	PROCLAMATION OF SALE UNDER THE NATIONAL LAND CODE / PEJABAT TANAH
I8	FORECLOSURE PROCEEDINGS STATUTORY NOTICE ISSUED
I9	PROCLAMATION OF SALE/OTHERS
ID	TRIBUNAL CASES

16 State Code

Code	Description
01	JOHOR
02	KEDAH
03	KELANTAN
04	MELAKA
05	NEGERI SEMBILAN
06	Pahang
07	PULAU PINANG
08	PERAK
09	PERLIS
10	SABAH
11	SARAWAK
12	SELANGOR
13	TERENGGANU
14	W.P. KUALA LUMPUR
15	WILAYAH PERSEKUTUAN LABUAN
16	WILAYAH PERSEKUTUAN PUTRAJAYA
00	OUTSIDE MALAYSIA

17 Country Code

Code	Description
AF	AFGHANISTAN
AL	ALBANIA
DZ	ALGERIA
AS	AMERICAN SAMOA
AD	ANDORRA
AO	ANGOLA
AI	ANGUILLA

Code	Description
AQ	ANTARTICA
AG	ANTIGUA AND BARBUDA
AR	ARGENTINA
AM	ARMENIA
AW	ARUBA
AU	AUSTRALIA
AT	AUSTRIA
AZ	AZERBAIJAN
BS	BAHAMAS
BH	BAHRAIN
BD	BANGLADESH
BB	BARBADOS
BY	BELARUS
BE	BELGIUM
BZ	BELIZE
BJ	BENIN
BM	BERMUDA
BT	BHUTAN
BO	BOLIVIA
BA	BOSNIA AND HERZEGOVINA
BW	BOTSWANA
BV	BOUVET ISLAND
BR	BRAZIL
IO	BRITISH INDIAN OCEAN TERRITORY
BN	BRUNEI DARUSSALAM
BG	BULGARIA
BF	BURKINA FASO
BI	BURUNDI
KH	CAMBODIA
CM	CAMEROON
CA	CANADA
CV	CAPE VERDE
KY	CAYMAN ISLANDS
CF	CENTRAL AFRICAN REPUBLIC
TD	CHAD
CL	CHILE
CN	CHINA
TW	CHINESE TAIPEI
CX	CHRISTMAS ISLAND
CC	COCOS (KEELING) ISLANDS
CO	COLOMBIA
KM	COMOROS
CG	CONGO
CD	CONGO, THE DEMOCRATIC REPUBLIC OF THE
CK	COOK ISLANDS
CR	COSTA RICA

Code	Description
CI	COTE D'IVOIRE
HR	CROATIA
CU	CUBA
CY	CYPRUS
CZ	CZECH REPUBLIC
DK	DENMARK
DJ	DJIBOUTI
DM	DOMINICA
DO	DOMINICAN REPUBLIC
TP	EAST TIMOR
EC	ECUADOR
EG	EGYPT
SV	EL SALVADOR
GQ	EQUATORIAL GUINEA
ER	ERITREA
EE	ESTONIA
ET	ETHIOPIA
XE	EUROPEAN COMMUNITY
FO	FAEROE ISLANDS
FK	FALKLAND ISLANDS (MALVINAS)
FJ	FIJI
FI	FINLAND
FR	FRANCE
GF	FRENCH GUIANA
PF	FRENCH POLYNESIA
TF	FRENCH SOUTHERN TERRITORIES
GA	GABON
GM	GAMBIA
GE	GEORGIA
DE	GERMANY
GH	GHANA
GI	GIBRALTAR
GR	GREECE
GL	GREENLAND
GD	GRENADA
GP	GUADELUOPE
GU	GUAM
GT	GUATEMALA
GG	GUERNSEY, C.I.
GN	GUINEA
GW	GUINEA-BISSAU
GY	GUYANA
HT	HAITI
HM	HEARD AND MCDONALD ISLANDS
VA	HOLY SEE (VATICAN CITY STATE)
HN	HONDURAS

Code	Description
HK	HONG KONG
HU	HUNGARY
IS	ICELAND
IN	INDIA
ID	INDONESIA
IR	IRAN (ISLAMIC REPUBLIC OF)
IQ	IRAQ
IE	IRELAND
IM	ISLE OF MAN
IL	ISRAEL
IT	ITALY
JM	JAMAICA
JP	JAPAN
JE	JERSEY, C.I.
JO	JORDAN
KZ	KAZAKHSTAN
KE	KENYA
KI	KIRIBATI
KP	KOREA, DEMOCRATIC PEOPLE'S REPUBLIC OF
KR	KOREA, REPUBLIC OF
KW	KUWAIT
KG	KYRGYZSTAN
LN	LABUAN INTERNATIONAL OFFSHORE FINANCIAL CENTRE
LA	LAO PEOPLE'S DEMOCRATIC REPUBLIC
LV	LATVIA
LB	LEBANON
LS	LESOTHO
LR	LIBERIA
LY	LIBYAN ARAB JAMAHIRIYA
LI	LIECHTENSTEIN
LT	LITHUANIA
LU	LUXEMBOURG
MO	MACAU
MK	MACEDONIA, THE FORMER YUGOSLAV REPUBLIC OF
MG	MADAGASCAR
MW	MALAWI
MY	MALAYSIA
MV	MALDIVES
ML	MALI
MT	MALTA
MH	MARSHALL ISLANDS
MQ	MARTINIQUE
MR	MAURITANIA
MU	MAURITIUS
YT	MAYOTTE
MX	MEXICO

Code	Description
FM	MICRONESIA (FEDERATED STATES OF)
MD	MOLDOVA, REPUBLIC OF
MC	MONACO
MN	MONGOLIA
MS	MONTSEERRAT
MA	MOROCCO
MZ	MOZAMBIQUE
MM	MYANMAR
NA	NAMIBIA
NR	NAURU
NP	NEPAL
NL	NETHERLANDS
AN	NETHERLANDS ANTILLES
NT	NEUTRAL ZONE BETWEEN SAUDI ARABIA AND IRAQ
NC	NEW CALEDONIA
NZ	NEW ZEALAND
NI	NICARAGUA
NE	NIGER
NG	NIGERIA
NU	NIUE
NF	NORFOLK ISLAND
MP	NORTHERN MARIANA ISLANDS
NO	NORWAY
OM	OMAN
OT	OTHERS
PK	PAKISTAN
PW	PALAU
PS	PALESTINIAN TERRITORY, OCCUPIED
PA	PANAMA
PZ	PANAMA CANAL ZONE
PG	PAPUA NEW GUINEA
PY	PARAGUAY
PE	PERU
PH	PHILIPPINES
PN	PITCAIRN
PL	POLAND
PT	PORTUGAL
PR	PUERTO
QA	QATAR
RE	REUNION
ME	REPUBLIC OF
RS	REPUBLIC OF SERBIA
RO	ROMANIA
RU	RUSSIAN FEDERATION
RW	RWANDA
SH	SAINT HELENA

Code	Description
KN	SAINT KITTS AND NEVIS
LC	SAINT LUCIA
PM	SAINT PIERRE AND MIQUELON
VC	SAINT VINCENT AND THE GRENADINES
WS	SAMOA
SM	SAN MARINO
ST	SAO TOME AND PRINCIPE
SA	SAUDI ARABIA
SN	SENEGAL
SC	SEYCHELLES
SL	SIERRA LEONE
SG	SINGAPORE
SK	SLOVAKIA
SI	SLOVENIA
SB	SOLOMON ISLANDS
SO	SOMALIA
ZA	SOUTH AFRICA
GS	SOUTH GEORGIA NAD SOUTH SANDWICH ISLANDS
ES	SPAIN
LK	SRI LANKA
SD	SUDAN
SR	SURINAME
SJ	SVALBARD AND JAN MAYEN ISLANDS
SZ	SWAZILAND
SE	SWEDEN
CH	SWITZERLAND
SY	SYRIAN ARAB REPUBLIC
TJ	TAJIKISTAN
TZ	TANZANIA, UNITED REPUBLIC OF
TH	THAILAND
TG	TOGO
TK	TOKELAU
TO	TONGA
TT	TRINIDAD AND TOBAGO
TN	TUNISIA
TR	TURKEY
TM	TURKMENISTAN
TC	TURKS AND CAICOS ISLANDS
TV	TUVALU
UG	UGANDA
UA	UKRAINE
AE	UNITED ARAB EMIRATES
GB	UNITED KINGDOM
US	UNITED STATES
UM	UNITED STATES MINOR OUTLYING ISLANDS
UY	URUGUAY

Code	Description
UZ	UZBEKISTAN
VU	VANUATU
VE	VENEZUELA
VN	VIETNAM
VG	VIRGIN ISLANDS, BRITISH
VI	VIRGIN ISLANDS, U.S
WF	WALLIS AND FUTUNA ISLANDS
EH	WESTERN SAHARA
YE	YEMEN
YU	YUGOSLAVIA
ZM	ZAMBIA
ZW	ZIMBABWE
XK	REPUBLIC OF KOSOVO

18 Error Message

There are 2 levels of Error message. The error message can be come from global which indicate the overall error. However, for subject level the error message is purely affected in the subject level only.

Global Level Error Message

XPath: /report/enq_report/header/enq_status

Code	Desc
1	SUCCESS
0	INVALID XML FORMAT Please contact CTOS Administrator. Please contact CTOS Support. Total Record is not numeric value! User Info not match! Password not match! Service is currently unavailable. Your request was encountered error. Please contact CTOS Support. Only individuals are allowed to be owners of a business! Business cannot be included as an Interest party for a company Business entity has to be requested first Mandatory for Business to be opted for SME Score Minimum one owner is required! Maximum 5 owners allowed! Maximum 10 directors/shareholders are allowed! The report Id: [REPORT_ID] is not found

Subject level Error Message

XPath: /report/enq_report/summary/enq_sum/enq_status

Code	Desc
2	MULTIPLE ENTITIES
1	SUCCESS
0	Analytical Product Service is Unavailable Analytical product requires to include CCRIS Analytics Product code is not allowed

	B2B Services Unavailable; Record within Retention Period not found Business Registration No required! Business Registration No's length exceeded 12 characters! Business Registration No's length exceeded 20 characters! Business cannot be included as an Interest party for a company Business entity has to be requested first CCRIS Service Unavailable CCRIS Services Unavailable CCRIS Services Unavailable; Record within Retention Period not found CCRIS is mandatory for Credit Insight to be generated Company Local No not found in SSM Company Local No required! Company Local No's length exceeded 7 characters Company has no access to CTOS with CCRIS report Consent found but record need update Consent found but the consent did not match the mobile number given in the request Consent revoke Consent web services return null Credit Insight Engine is unavailable Credit Insight is disabled for your account Credit Insight is only available for party type I DistCode length exceeded 50 characters EndDate is in the future; Record within Retention Period not found Entity Not Found in SSM FICO Web Service failed to get score! Failed to generate Section E1 Failed to generate Section E2 Failed to start database connection Foreign company Local No's length exceeded 20 characters IC or NIC required! IC's length exceeded 8 characters Include_Angkasa tag is either 0 or 1 value! Include_Ccris tag is either 0 or 1 value! Include_Ctos tag is either 0 or 1 value! Include_DCheque tag is either 0 or 1 value! Include_Le tag is either 0 or 1 value! Include_Sfi tag is either 0 or 1 value! Include_Ssm tag is either 0 or 1 value! Include_Trex tag is either 0 or 1 value! Include_ccris_supp tag is either 0 or 1 value! Include_fico tag is either 0 or 1 value! Income Estimator Service is Unavailable Income Estimator is only available for Individuals Invalid Analytics Product code Invalid FI Code LLP Registration No required! LLP Registration No's length exceeded 13 characters Local No is allowed numeric only Mandatory for Business to be opted for SME Score Maximum 5 owners allowed! Maximum 10 directors/shareholders are allowed! Minimum one owner is required! MphoneNo length exceeded 255 characters NIC's length exceeded 13 characters No Consent No Consent - [error message returned by ctosid's consent] Only individuals are allowed to be owners of a business! Purpose Code cannot be null or Empty value
--	---

	Ref No's length exceeded 50 characters Request Name cannot be null or Empty value Request Type cannot be null or Empty value SME FICO Web Service failed to get score! SSM Live is not available for East Malaysia Business SSM Retention is only available for party type B and C! SSM Service Unavailable Service Unavailable The analytical product opted is not available for this party type. The report is being disputed by the subject and will be suppressed pending completion of the investigation User has no access to CTOS with CCRIS report XML DCheqs XML Format include_ci tag is either 0 or 1 value! include_etr_plus tag is either 0 or 1 value!
3	This is not an existing customer for this FI

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